Chapter 5: Metabolism & Energy

- What is energy?
- Chemical reactions capture & release energy
- What kind of energy is used in cells?
- Chemical reactions are regulated by enzymes
- How do we get usable energy on Earth?
- How do we release energy trapped in molecules?

Corresponds with OpenStax Biology 2e Chapters 6, 7, and 8

Bioenergetics

- **Definition:** Quantitative study of the energy transductions that occur in living cells and study of the nature and function of the chemical processes underlying the transductions

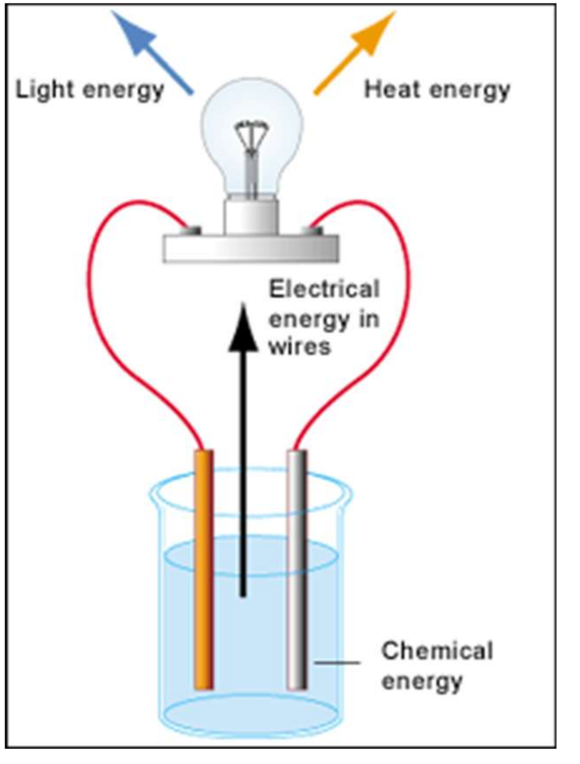
- Simple Definition: The study of energy in living systems (environments) and the organisms (plants and animals) that utilize them

- The goal of bioenergetics is to describe how living organisms acquire and transform energy in order to perform biological work

Figure: https://www1.curriculum.edu.au/sciencepd/energy/elec_activity.htm

What is energy?

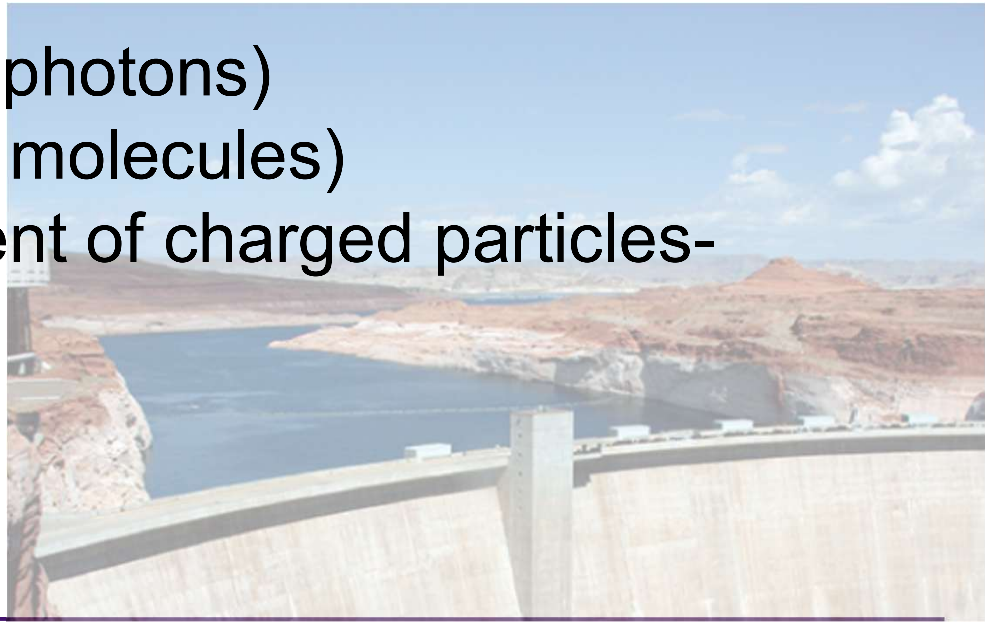
- **Energy** = the capacity to do *work*
  - To work means to move something – all of life depends on rearranging atoms & trafficking molecules, so life requires movement / work / energy
    - Energy comes in many different types: heat, light, electricity, chemical, radiation – all result from movement / work



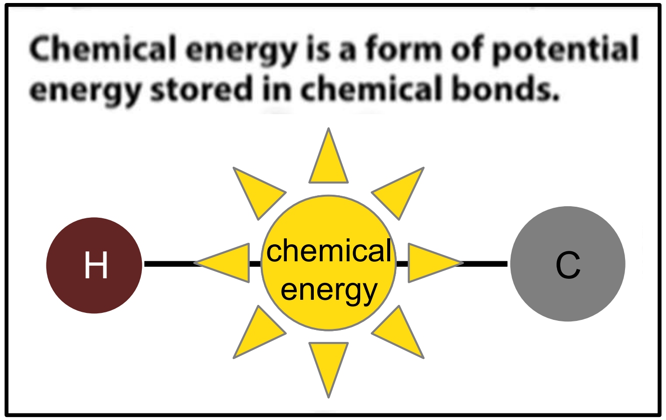
All types of energy can be in one of two forms

- **Potential energy** = stored energy
  - e.g. an unburned molecule of gasoline contains potential energy that can be used by an engine
  - e.g., water

- **Kinetic energy** = energy of motion or movement, used to do work
  - e.g. light (movement of photons)
  - e.g. heat (movement of molecules)
  - e.g. electricity (movement of charged particles- electrons, ions)
  - e.g. water



- We **store chemical energy** (potential) by building bonds in molecules
  - e.g. gasoline, oil, & coal have energy in their chemical bonds
  - e.g. sugars & fats have energy stored in their chemical bonds

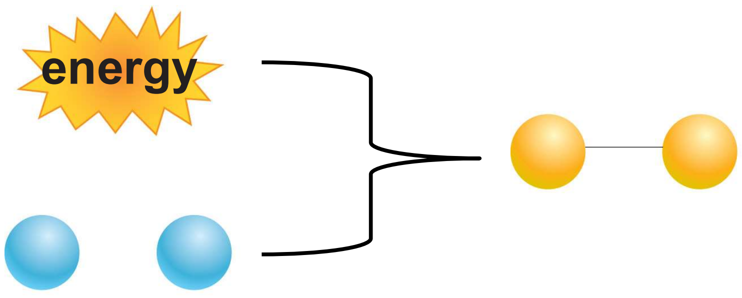


- We **release chemical energy** (kinetic – movement) by breaking the bonds
  - e.g. we power engines by breaking the bonds in gasoline molecules
  - e.g. we power our cells by breaking the bonds in sugars & fats

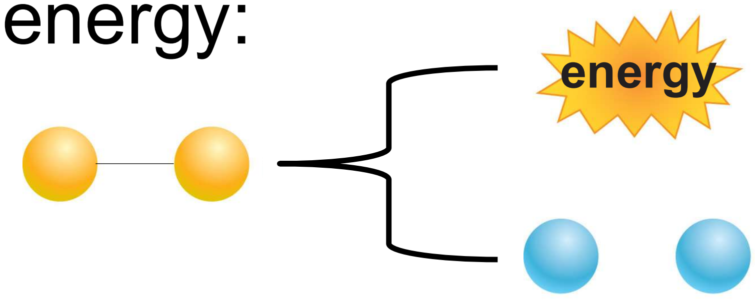
- <u>Energy can change forms:</u>
  - It can go from kinetic → potential *(storing energy)*
    - *we would **build** chemical bonds in a molecule to store energy*
  - It can go from potential → kinetic *(releasing energy)*
    - *we would **break** chemical bonds in a molecule to release energy*
  - Both of these are examples of **chemical reactions** = processes that build or break chemical bonds between atoms in a molecule

Chemical reactions capture & release energy

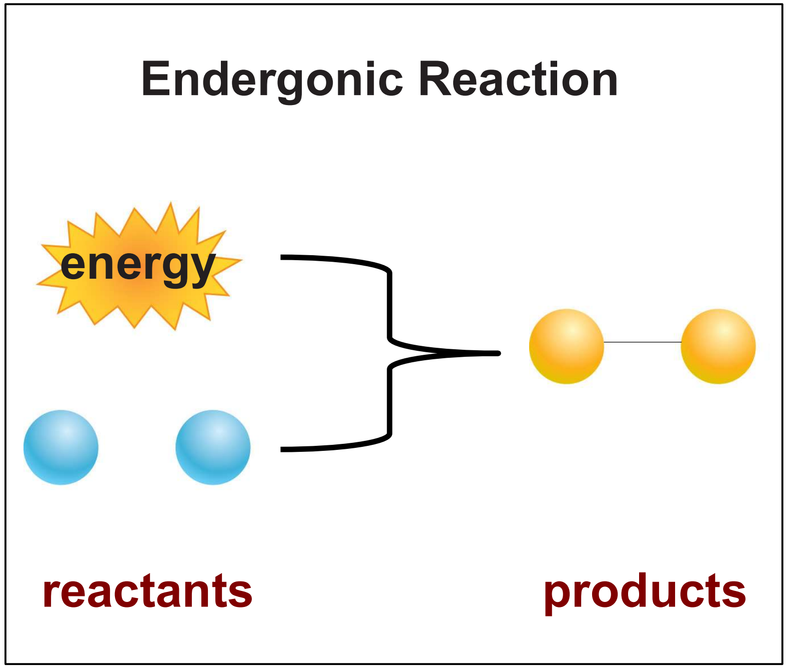
- There are 2 kinds of chemical reactions
  - The reaction that stores energy: kinetic → potential = an **endergonic** reaction

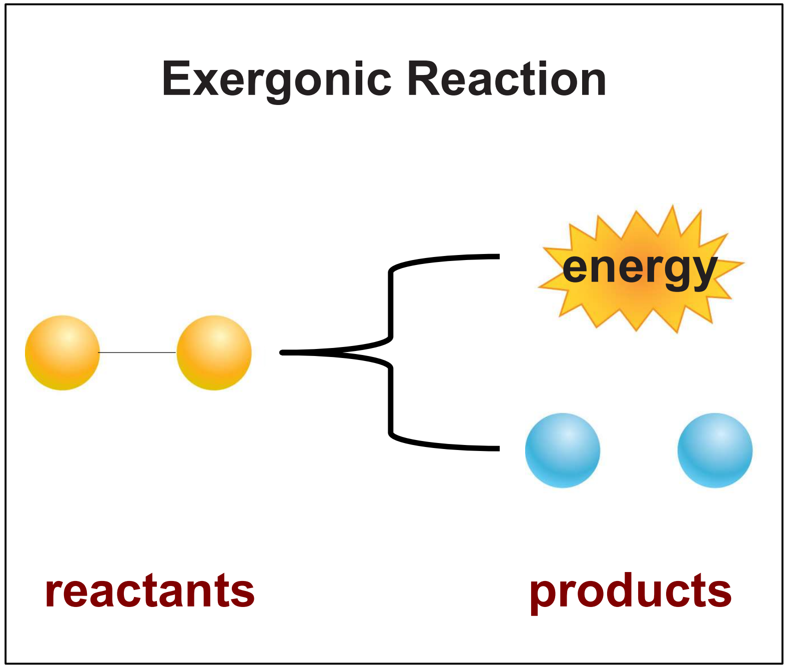


  - The reaction that releases energy: potential → kinetic = an **exergonic** reaction

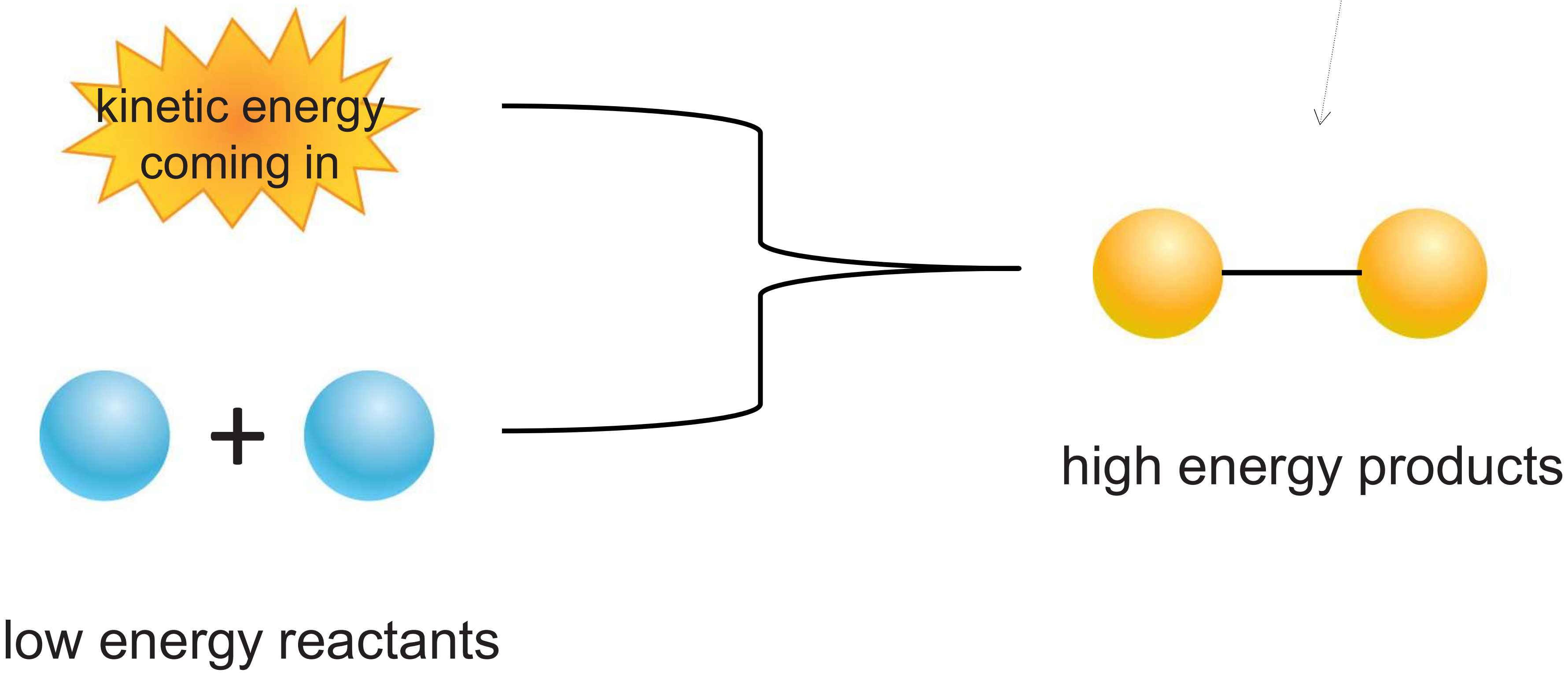


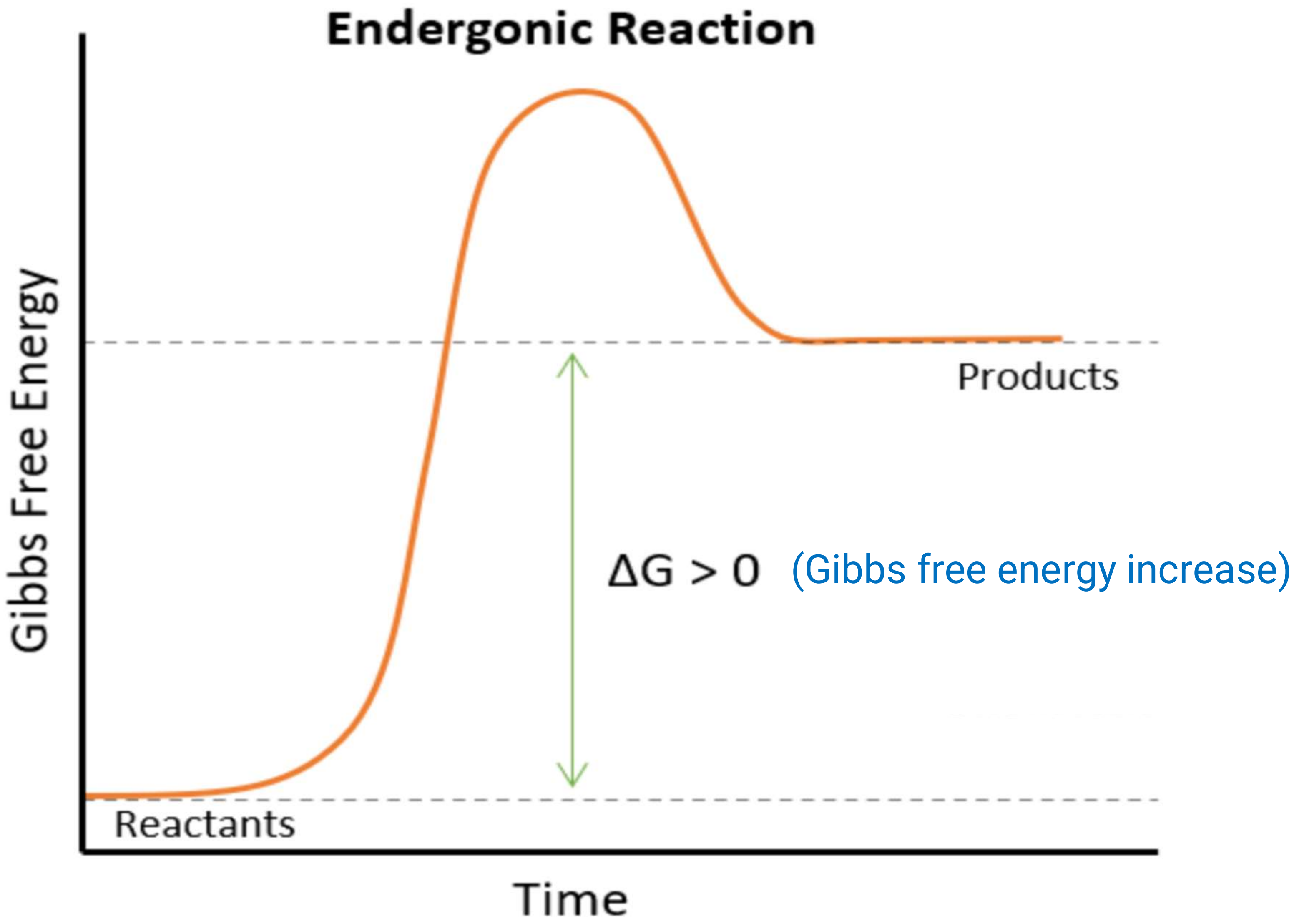
- In both kinds of chemical reactions we start with **reactants** & end with **products**





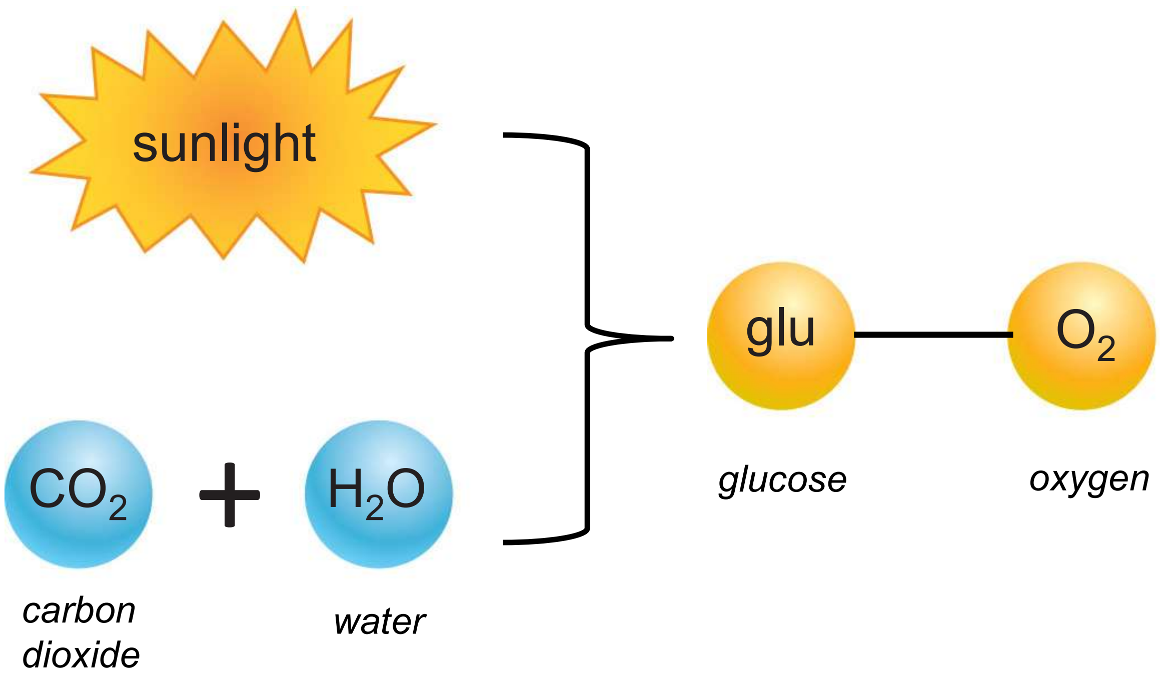
- <u>Endergonic reactions</u>
  - The reactants are low energy
  - Kinetic energy comes in & is stored in bonds
  - The products are high energy

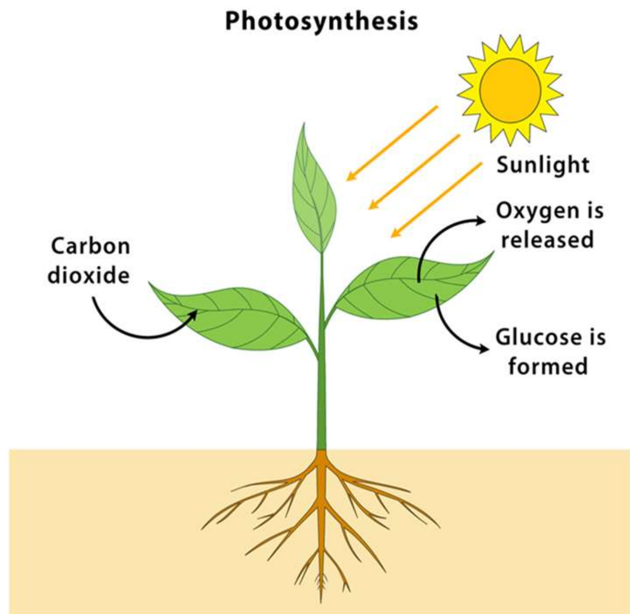



Endergonic Reaction
Gibbs Free Energy
Products
ΔG > 0
(Gibbs free energy increase)
Reactants
Time

- <u>Example endergonic reaction = photosynthesis</u>

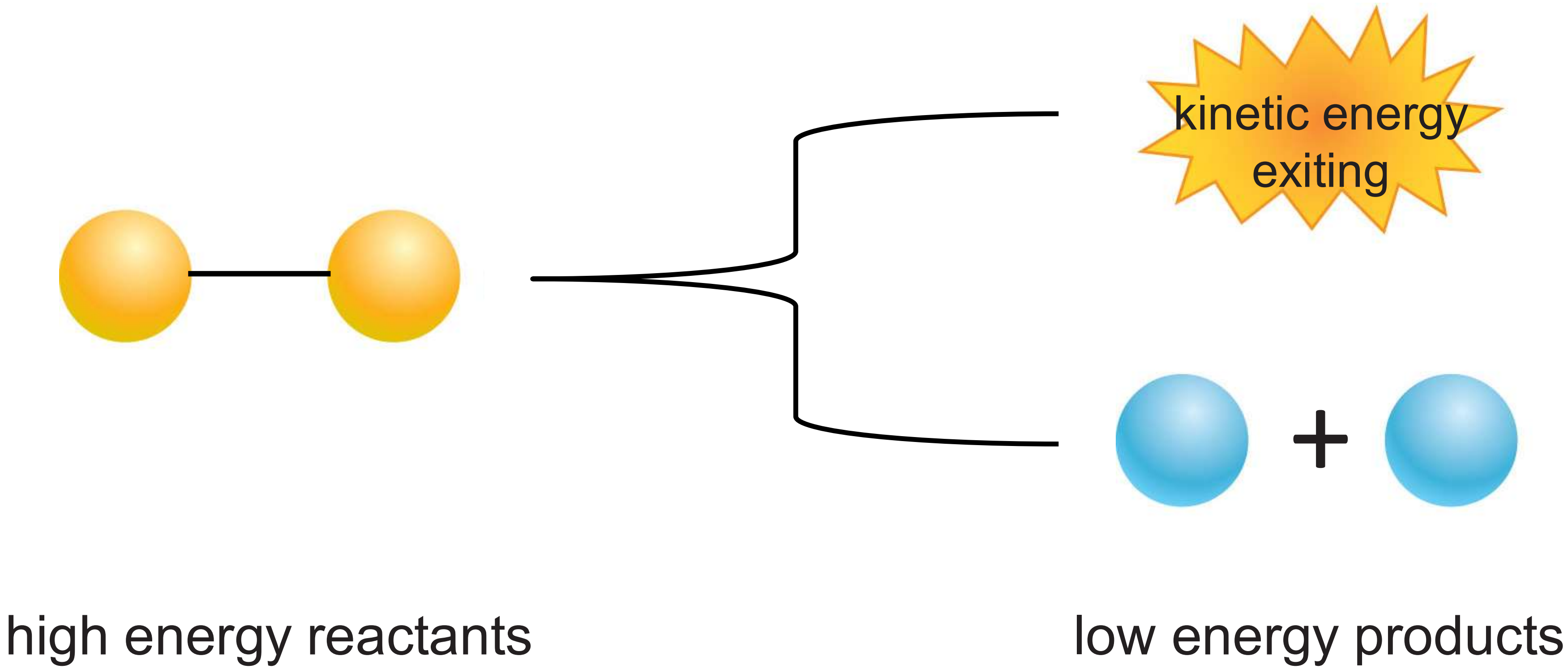
  – kinetic energy of sunlight coming in + water & CO_2 (low energy reactants) = sugar & O_2 (high energy products)

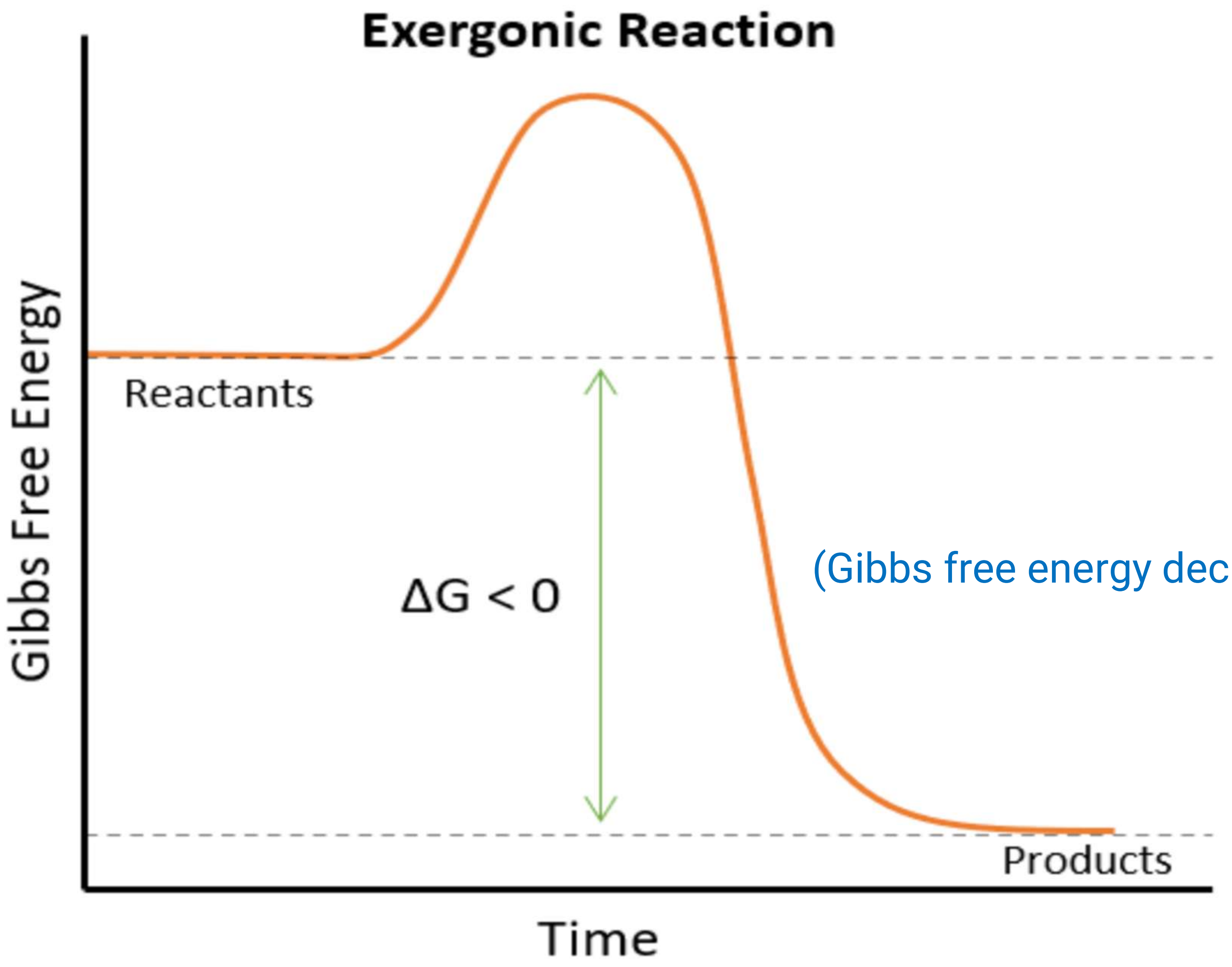




https://www.chemistrylearner.com/chemical-reactions

- <u>Exergonic reactions</u>
  - The reactants are high energy
  - Bonds are broken & kinetic energy comes out
  - The products are low energy

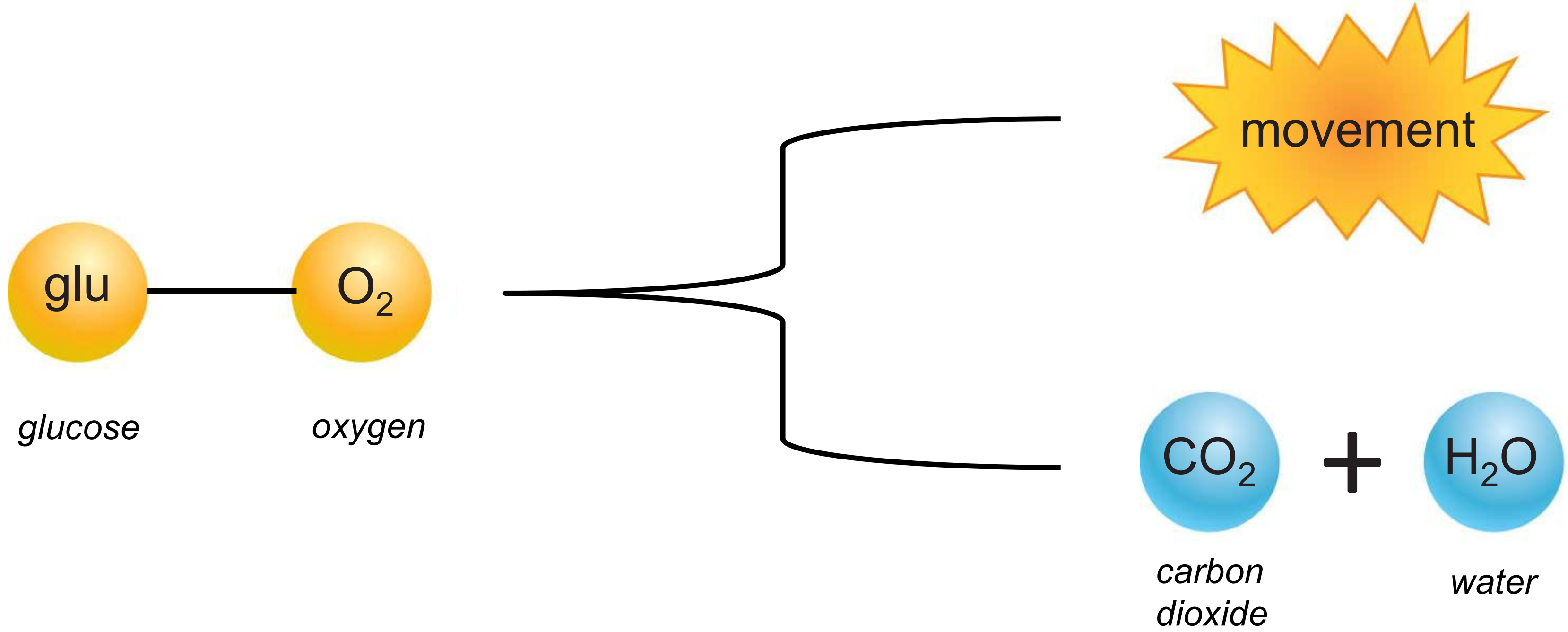




(Gibbs free energy decrease)

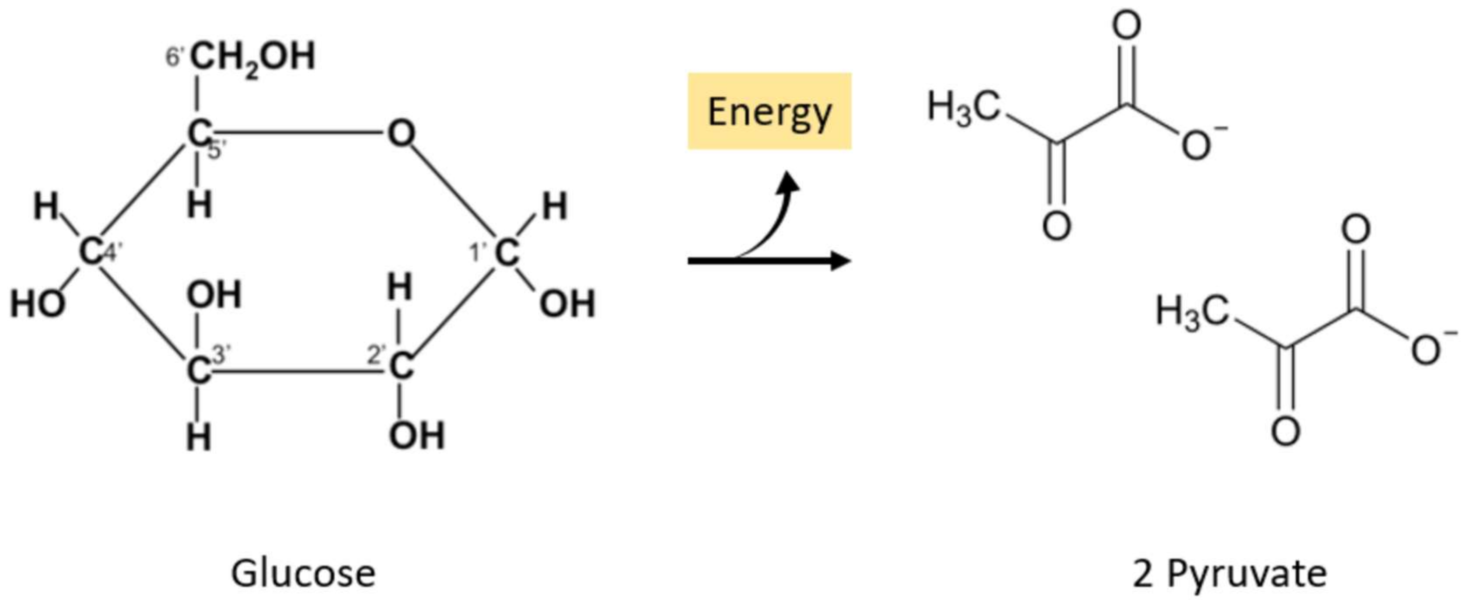
- Example exergonic reaction = glucose breakdown

  – as a glucose molecule is broken down, the sugar combines with O_2 (high energy reactants) to produce CO_2 & water (low energy products), releasing kinetic energy



- <u>Exergonic reaction example： Glycolysis</u>

Glycolysis is a biochemical pathway that starts with the sugar glucose ($C_6H_{12}O_6$), breaks it down into smaller molecules, and generates useful energy for a living system.



- Exergonic reaction example： Glucose breakdown

One molecule of glucose transforms chemically in ten distinct reactions or steps. These steps produce energy (ATP) and create important molecules. One of these important molecules is pyruvate, the end product of glycolysis. Pyruvate is special because it is a starting molecule for other energy-producing processes.

▪ Exergonic reaction example: Glycolysis in life

Glycolysis is a biochemical pathway that starts with the sugar glucose ($C_6H_{12}O_6$), breaks it down into smaller molecules, and generates useful energy for a living system.

- <u>Exergonic reactions are not efficient</u>
  - the conversion of potential energy to kinetic energy is not 100% efficient: waste product is **heat**

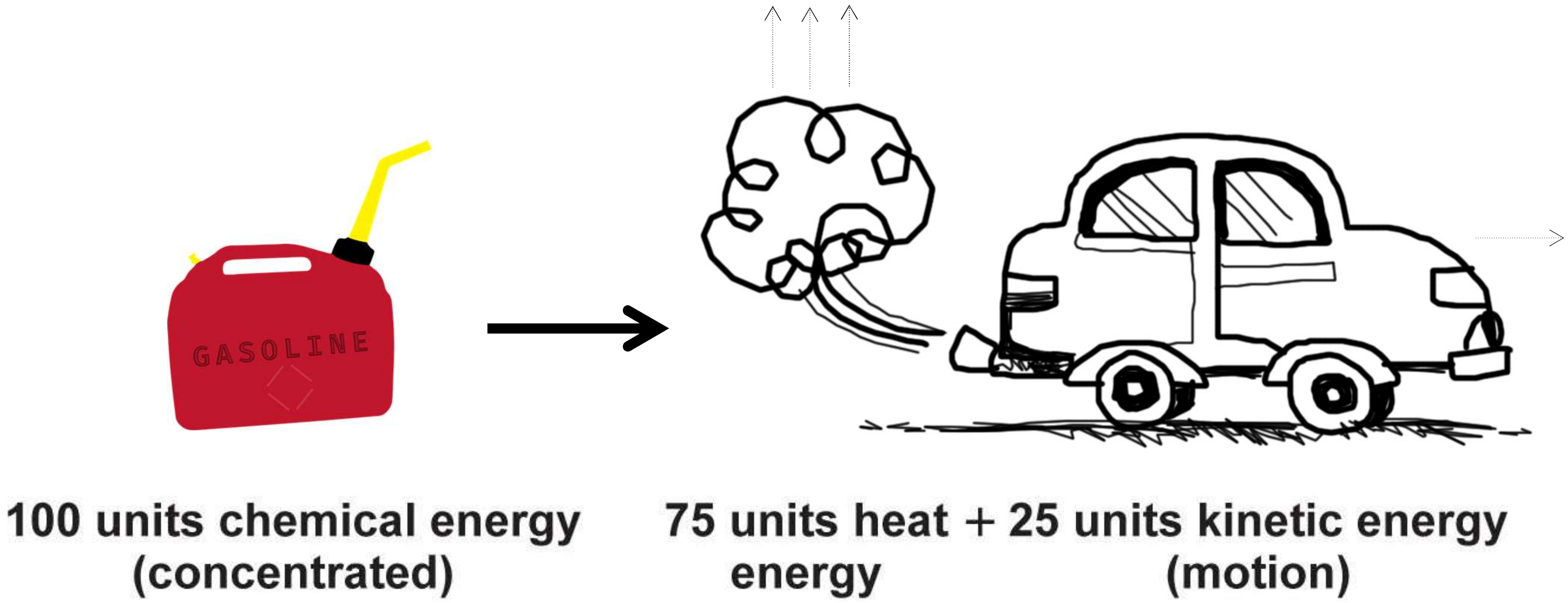


Figure: https://www.maxpixel.net/Flammable-Red-Danger-Canister-Container-Gas-Fuel-4161953

Figure: https://www.maxpixel.net/Exhaust-Gases-Doodle-Auto-1388117

Exergonic vs endergonic processes overview

	Exergonic	Endergonic
What happens to the energy?	Released	Absorbed
ΔG	Negative	Positive
Energy of reactants	Higher than products	Lower than products
Input of energy	Not required	Required
Spontaneity	Spontaneous/favorable	Unspontaneous/unfavorable
Exothermic or endothermic	Exothermic	Endothermic

Ice melting: endergonic or exergonic reaction?

When ice melts into water, it uses heat energy from its surroundings

→ Endergonic process

Sodium+Water: endergonic or exergonic reaction?

$$2Na_{(s)} + 2H_2O \rightarrow 2NaOH_{(aq)} + H_{2(g)}$$

Exergonic vs endergonic processes overview

Practice Problems

Problem 1: Cellular respiration is a metabolic pathway that breaks down glucose to produce ATP. What type of reaction is this?

Exergonic reaction

Problem 2: What does it mean when the ΔG of a reaction is equal to zero?

This means that the chemical reaction is in equilibrium. When this happens, the relative concentration of products and reactants does not change. This happens because the products and reactants are converting into each other at an equal rate.

Problem 3: If the energy of the reactants is less than the energy of the products, what type of reaction is it?

Endergonic reaction

What kind of energy is used in cells?

- All living things are powered by breaking apart **glucose** molecules
  - *Remember: breaking a molecule apart = hydrolysis*

– However: the energy in glucose can't be used directly – must first transfer it to a middle man: **energy-carrier molecule**

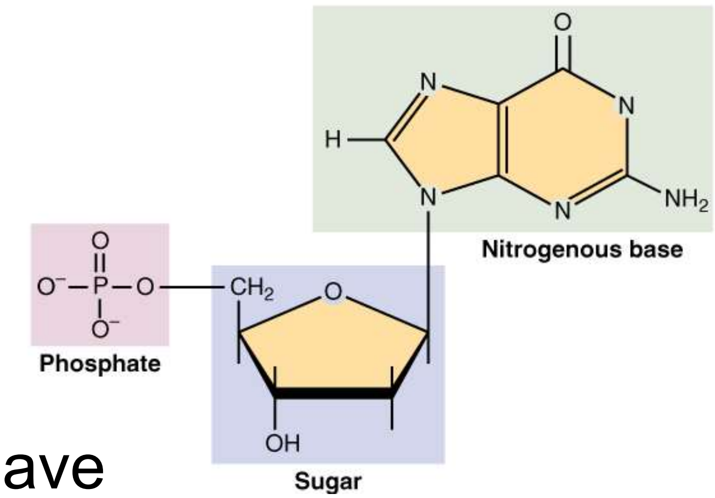
= molecules that carry energy from breaking apart glucose to anywhere in the cell that needs it

- most common energy-carrier molecule = **ATP**

adenosine triphosphate

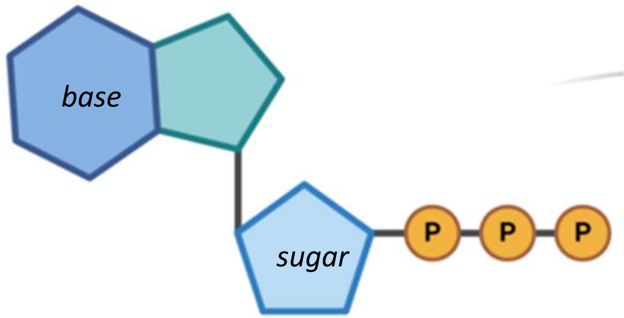
ATP is the most common energy-carrier molecule

- Remember from chapter 2 :
  - ATP carries energy (from breaking down carbohydrates like glucose) around the cell to where it's needed

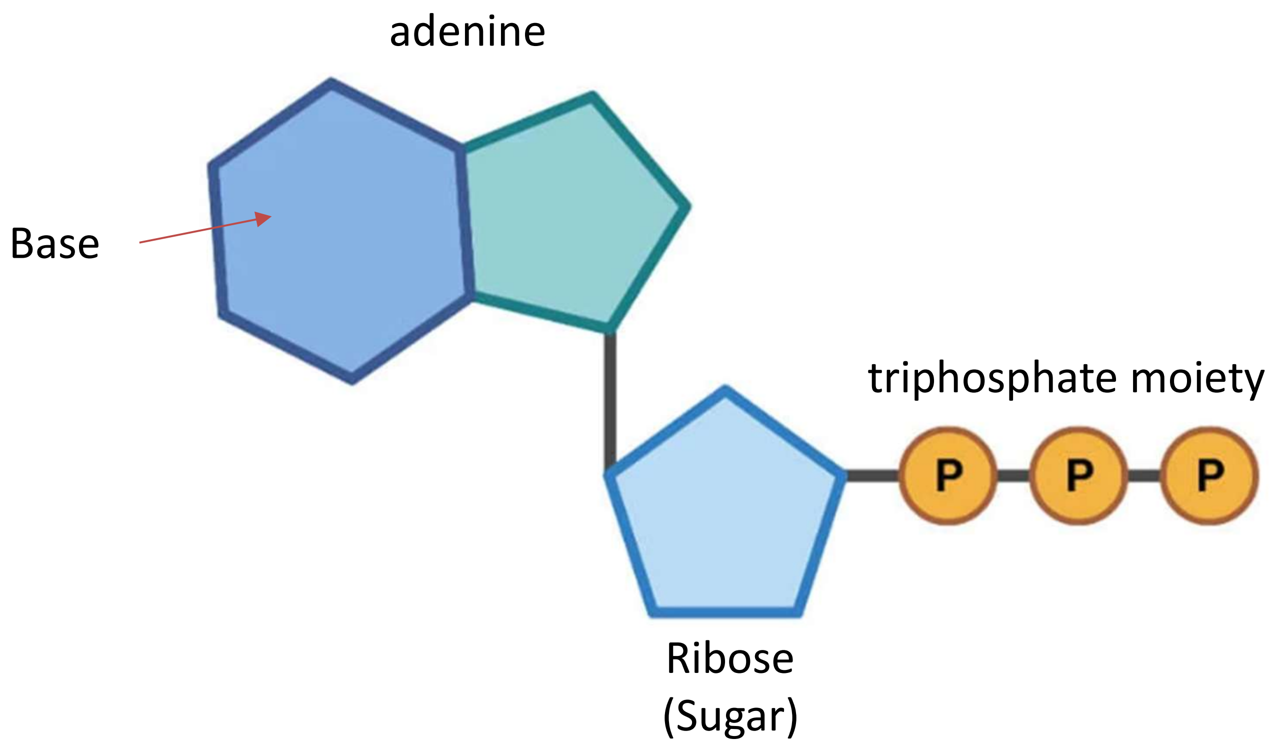


  - ATP = a nucleotide, which always have at least 1 phosphate, a sugar, & a base

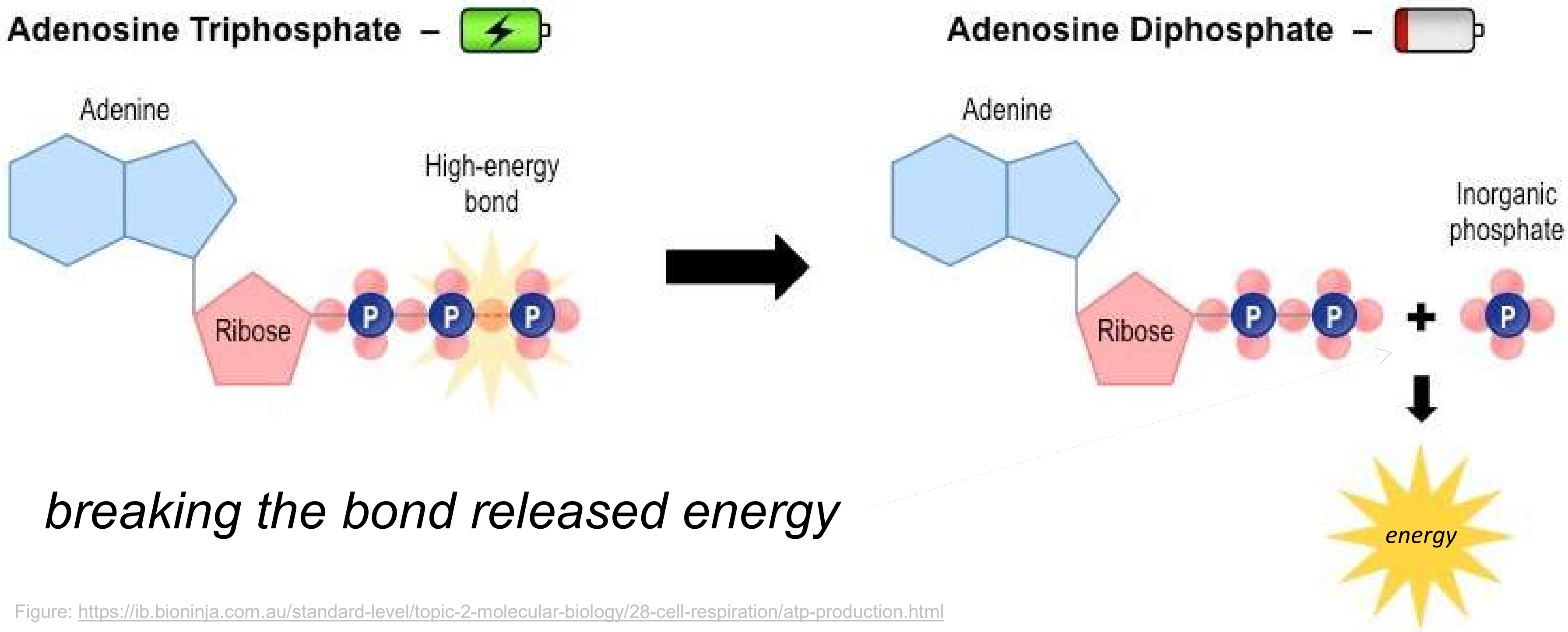
  - **ATP = adenosine triphosphate**: 3 phosphates help the molecule carry energy



ATP (Adenosine Triphosphate)



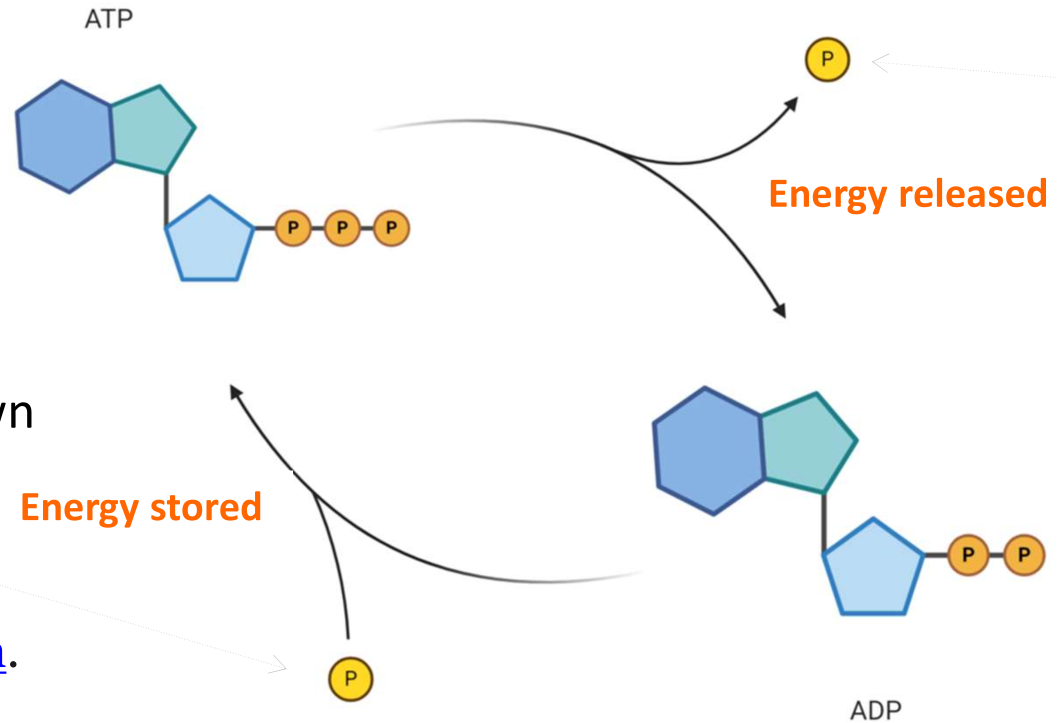
- To <u>release energy</u> from ATP (to fuel our cells), we must <u>break a bond</u> in the ATP molecule
  - The bond we break is between 2 phosphates
    - leaves **ADP** *(adenosine diphosphate)* & a **free phosphate**



breaking the bond released energy

Figure: https://ib.bioninja.com.au/standard-level/topic-2-molecular-biology/28-cell-respiration/atp-production.html

- ADP & a free phosphate can be reconnected again (with incoming kinetic energy from glucose breakdown)
  = ATP, which is charged up again with energy
    - In this way, ATP acts like a rechargeable battery



Energy is released as a phosphate is broken off from ATP.

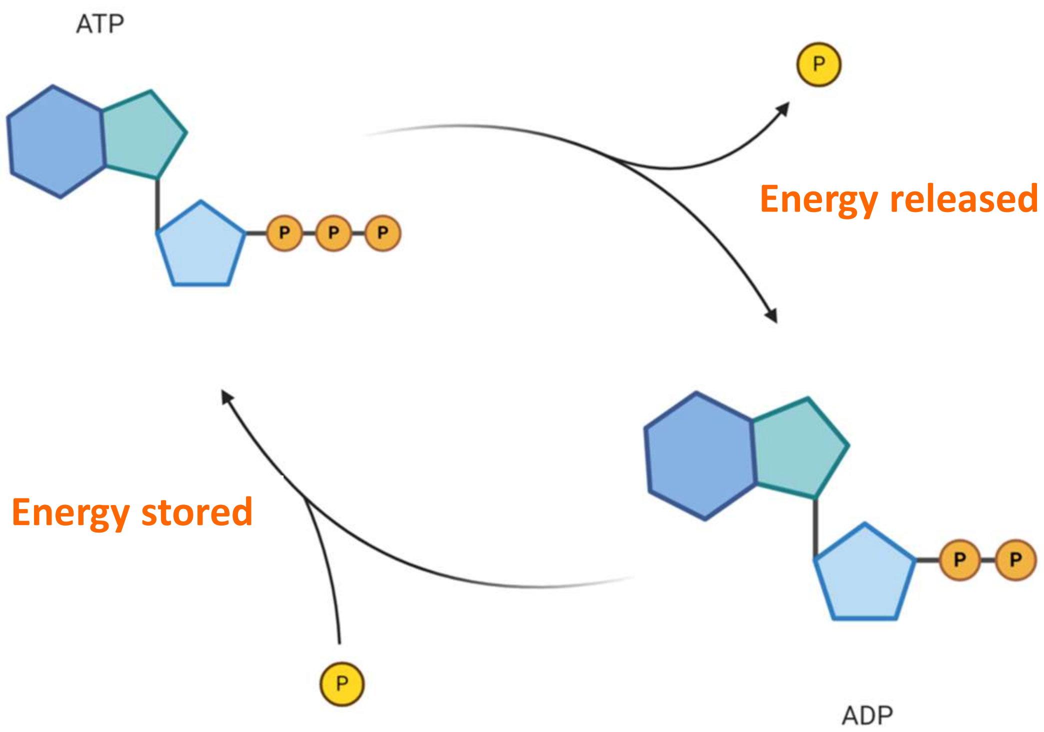
An input of energy from the breakdown of sugar/glucose attaches ADP to P.

[cellular respiration](https://sml.snl.no/ATP).

Figure: https://sml.snl.no/ATP

- 1 molecule of ATP is recycled thousands of times
  - Building a bond between uncharged ADP & P = storing energy – endergonic reaction
  - Breaking the bond in ATP (to make ADP & P) = releasing energy – exergonic reaction

An input of energy from the breakdown of food attaches ADP to P: **endergonic**



Energy is released as a phosphate is broken off from ATP: **exergonic**

Figure: https://sml.snl.no/ATP

Functions of ATP

1. ATP plays a significant role in anabolic reactions by providing energy for bone formation or breaking. It is the primary source of energy for cellular reactions and processes. Energy is stored and transported in the form of ATP inside living cells. All other forms of chemical energy in the cell are converted to ATP before use.
2. Vital processes like muscle contraction-relaxation, cellular movements, impulse transmission, heart pumping, blood circulation, etc., require ATP hydrolysis as fuel.
3. ATP is used as the energy source for transporting molecules in and out of the cell during active transport mechanisms.
4. ATP acts as an intracellular reserved source of energy.
5. ATP is involved in intracellular signaling processes. They serve as the substrate for kinases for phosphate transfer, adenylate cyclase enzymes, etc. ATP is converted to cAMP (cyclic AMP), which acts as secondary signaling molecules during intracellular signaling processes.
6. ATP is also involved in extracellular signaling and neurotransmission. During the purinergic signaling process, ATP is used for cell-to-cell communication. It also serves as a neurotransmitter in several neural signaling processes.

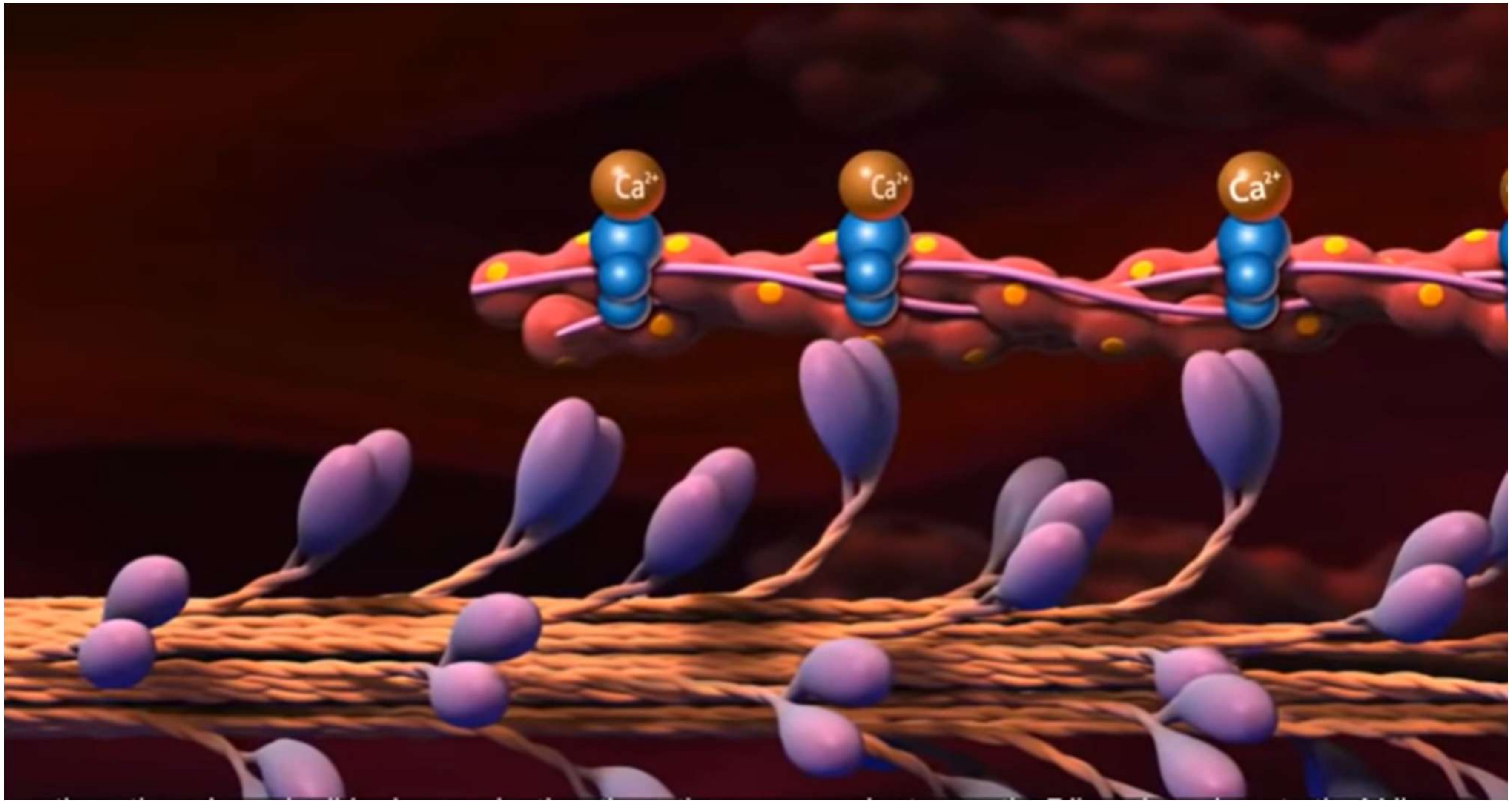
Figure: https://twitter.com/amoebasisters/status/1438293316240646148?s=21

Functions of ATP

7. ATP is required for the biosynthesis of DNA and RNA molecules. DNA gyrase of prokaryotes or DNA topoisomerase II requires ATP in form of dATP (deoxyribonucleotide adenosine triphosphate).
8. It is also involved in protein synthesis reactions by activating aminoacyl – tRNA synthetase enzymes.
9. Several ATP binding cassette transporters (ABC transporters) are present in cell membranes which use the energy of ATP binding and hydrolysis for cellular transportation like the uptake of vitamins, metal ions, biosynthetic precursors etc and efflux of lipids, drug residues, sterols, etc.
10. Injectable ATPs are used as diagnostic and therapeutic drugs for certain cardiac disorders (cardiac bradyarrhythmias).
11. ATP is found to act as a biological hydrotrope. ATP can hinder the thermal aggregation of proteins and the solubility of proteins.
12. ATP is also being studied for its anti-aging properties and is used in anti-aging drugs.

Figure: https://twitter.com/amoebasisters/status/1438293316240646148?s=21

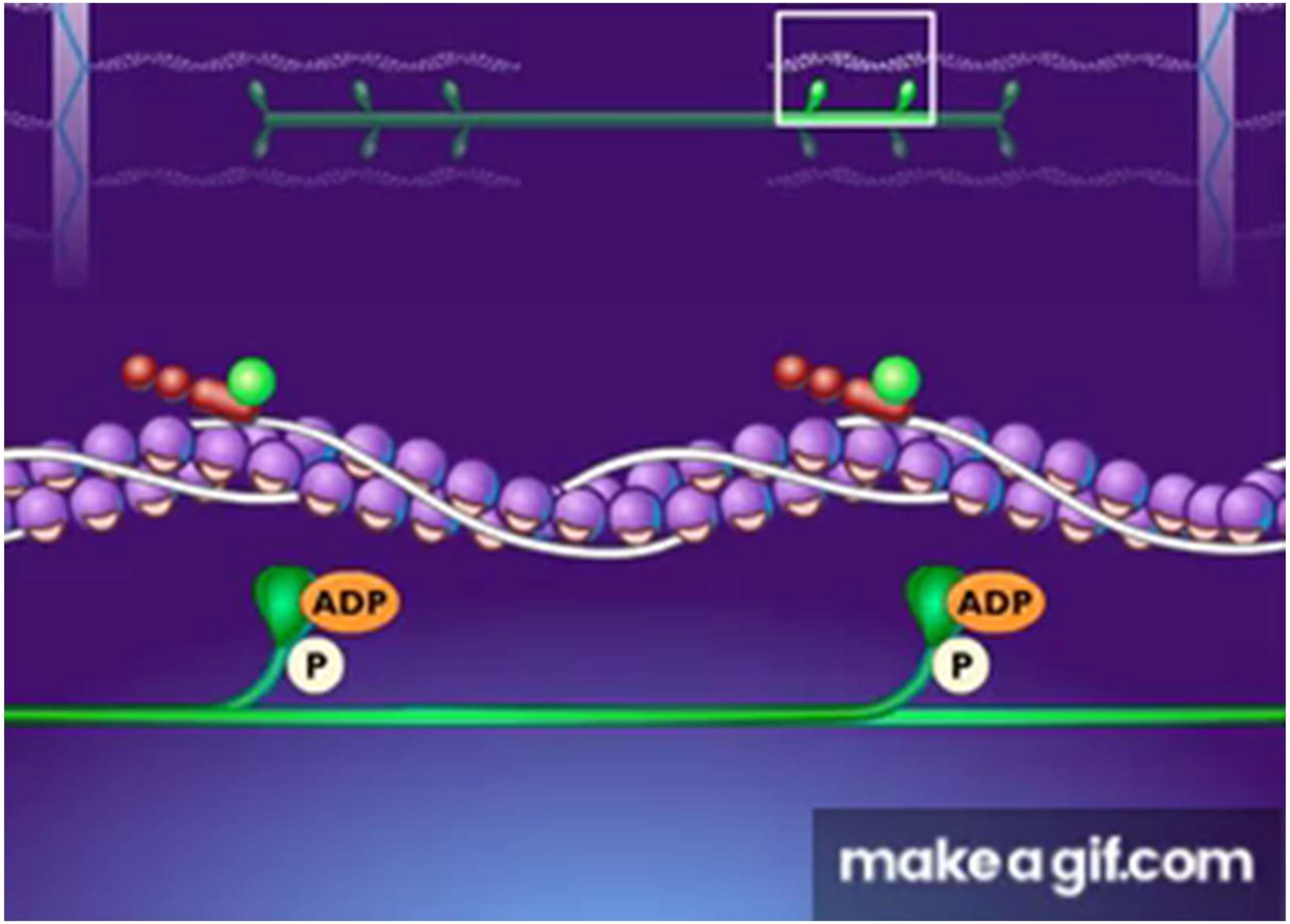
ATP hydrolysis as fuel in muscle contraction-relaxation



[Muscle Contraction Process Molecular Mechanism 3D Animation - video Dailymotion]

https://www.dailymotion.com/video/x3mshxf

ATP hydrolysis as fuel in muscle contraction-relaxation



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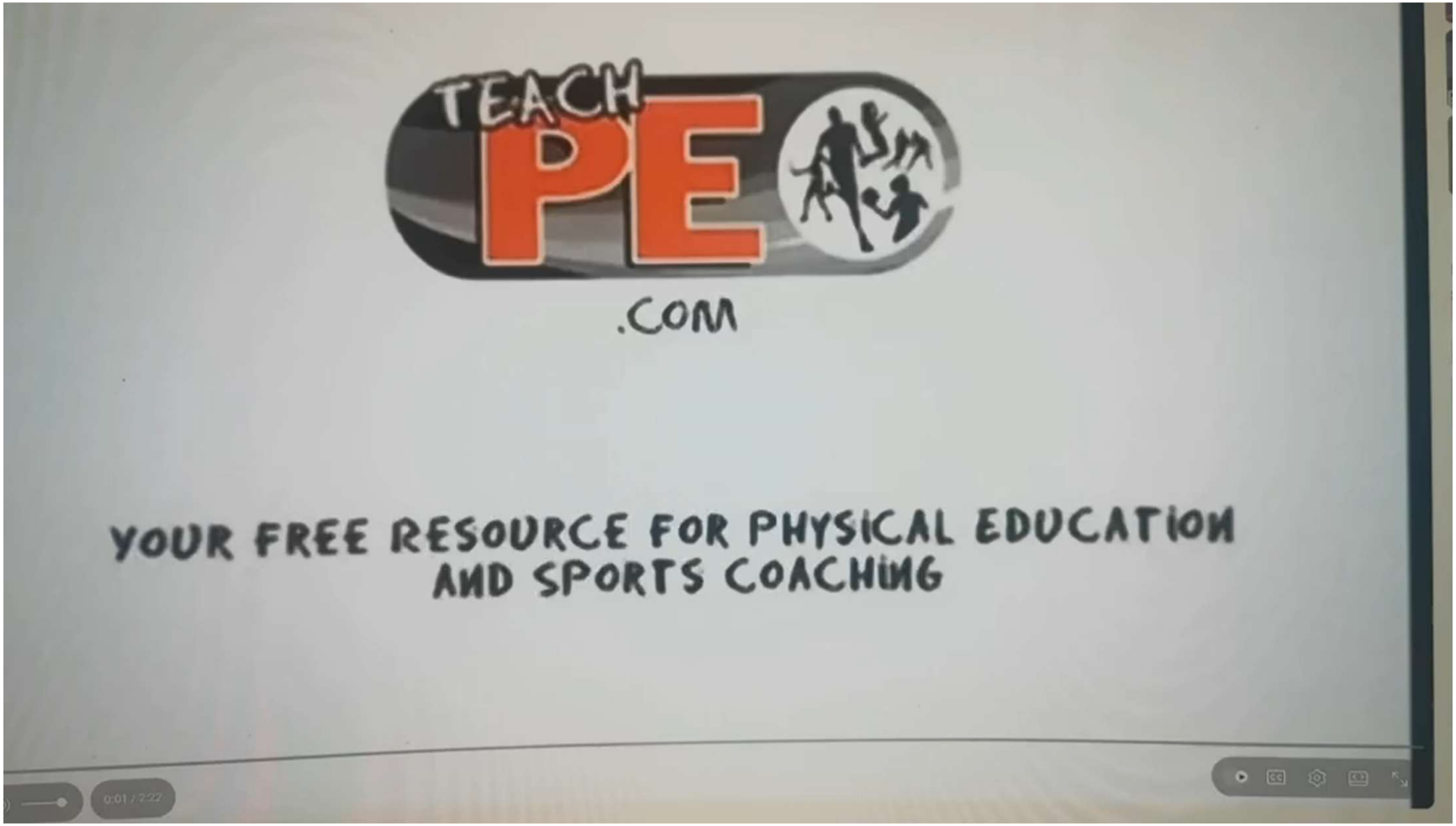
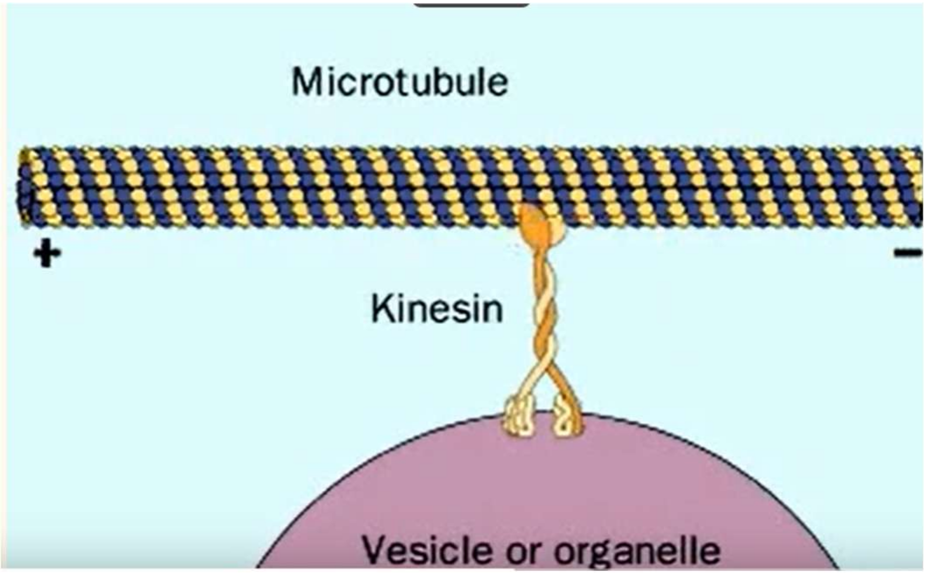
General Reading

ATP hydrolysis as fuel in muscle contraction-relaxation

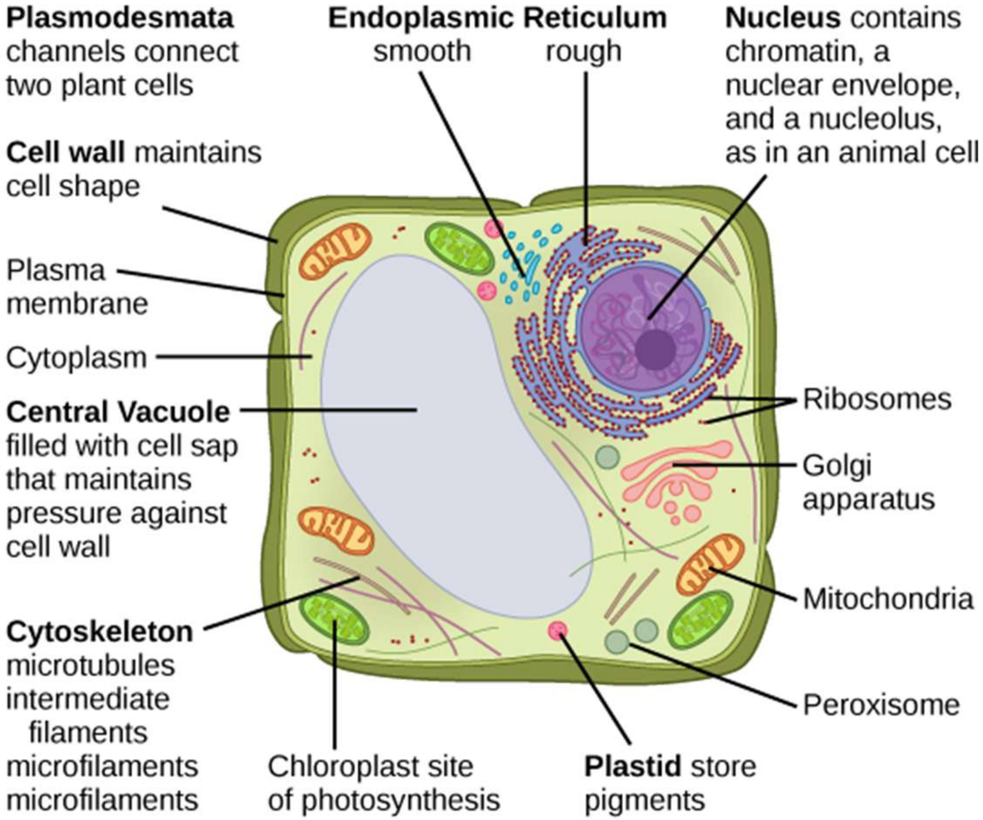


ATP hydrolysis as fuel in kinesin moving along a microtubule



We have learned:
Organelles = compartments that carry out specialized functions (little organs), e.g. mitochondria, chloroplasts

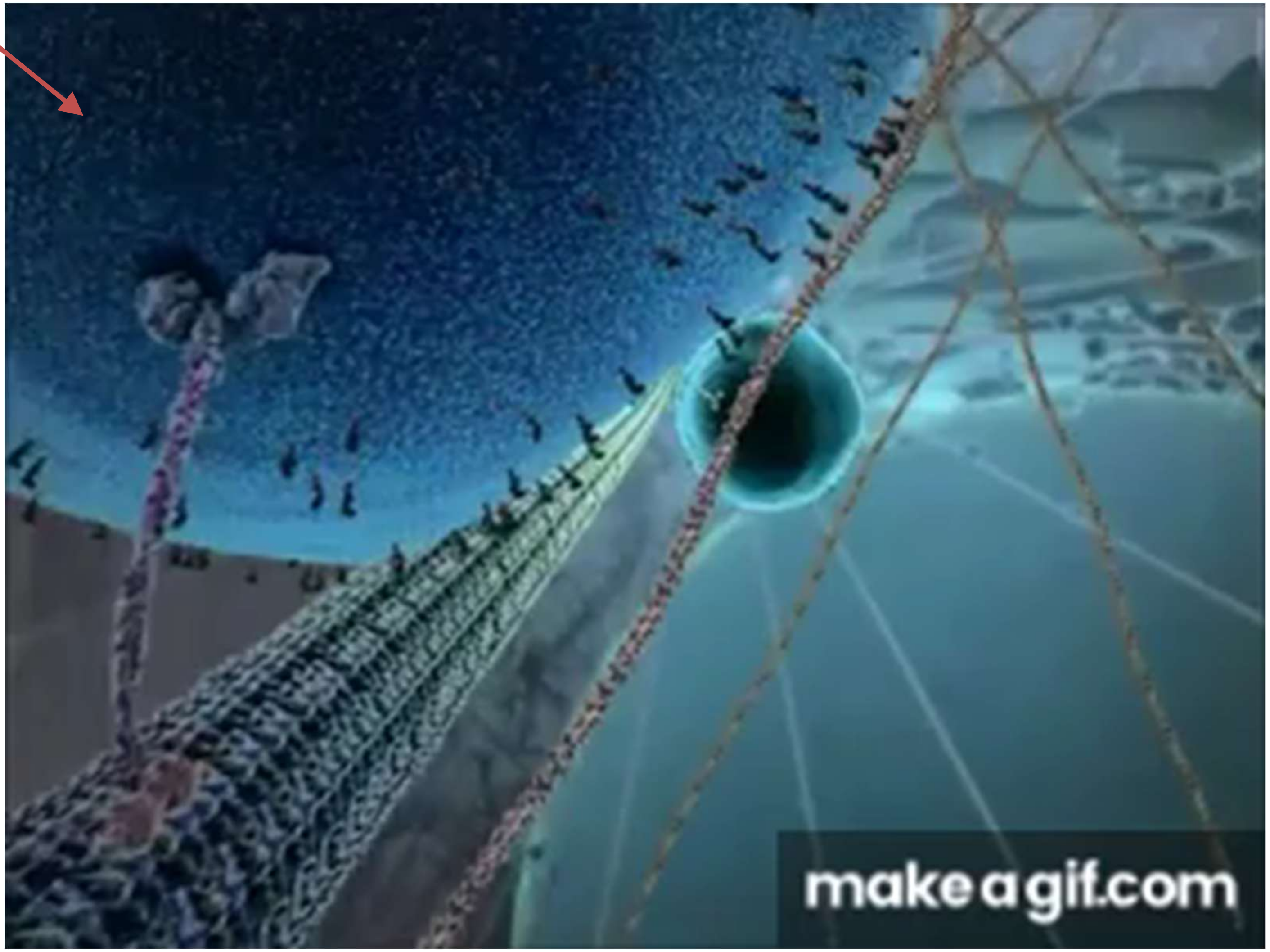
Vesicles = compartments formed by a lipid bilayer, e.g., vacuoles, exosomes



General Reading

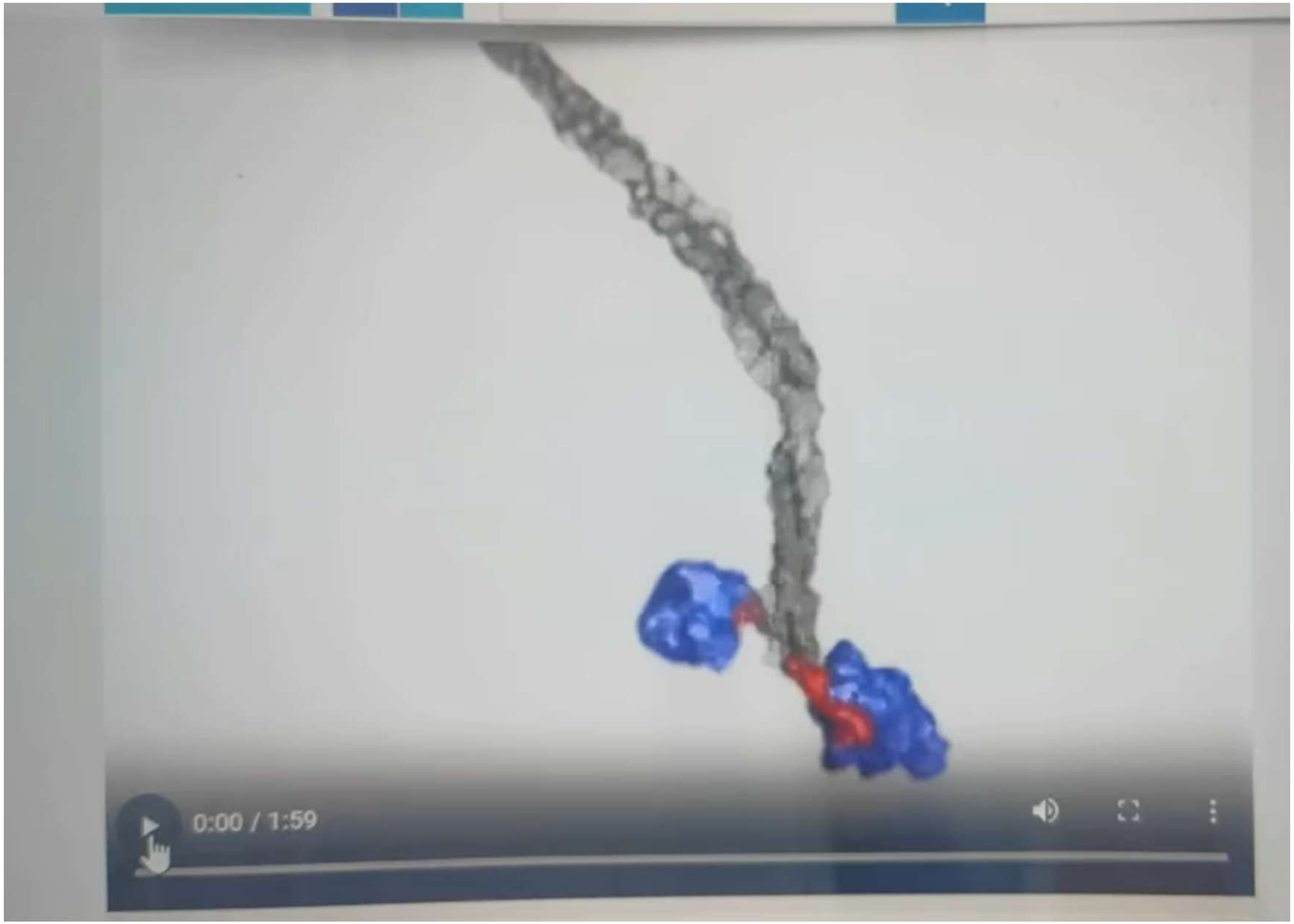
ATP hydrolysis as fuel in kinesin moving along a microtubule

Organelles



~8nm a step !

ATP hydrolysis as fuel in kinesin moving along a microtubule



~8nm a step！

https://www.dnatube.com/video/376/Kinesin

Enzymes

- Enzymes are biological catalysts, primarily proteins, that speed up chemical reactions in living organisms without being consumed.

- They are crucial for processes like digestion, metabolism, and building cellular components.

- There are many types of enzymes, each highly specific to a particular reaction and substrate molecule, and their function is influenced by factors like temperature and pH.

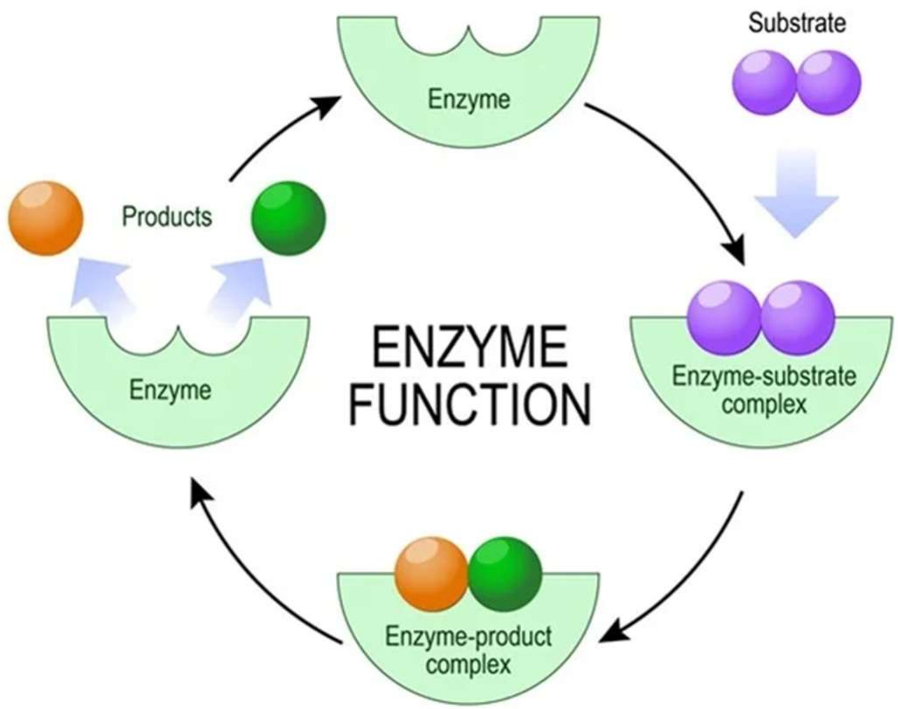


Figure: https://commons.wikimedia.org/wiki/File:212_Enzymes-01.jpg

Chemical reactions are regulated by enzymes

- Chemical reactions need a certain amount of energy to get started: **activation energy**
  - with no help, it can take too long for a chemical reaction to get over its activation energy: this means life can stop (death)

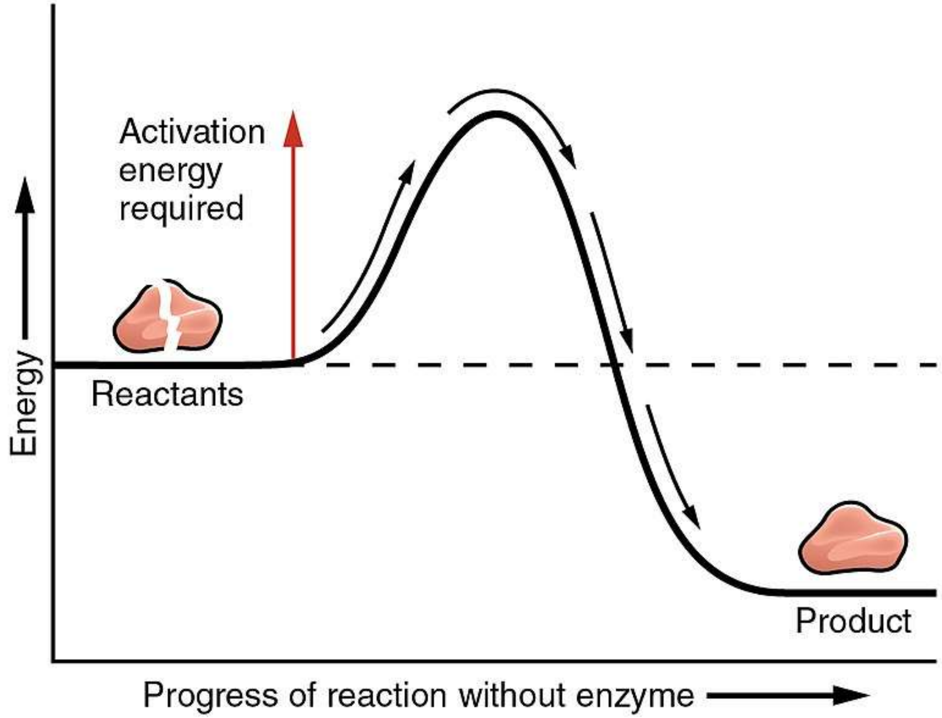


Figure: https://commons.wikimedia.org/wiki/File:212_Enzymes-01.jpg

- However, there are protein molecules that can give chemical reactions a “push” to help get them overcome their activation energy: **enzymes**
  – reactions can occur fast enough = staying alive

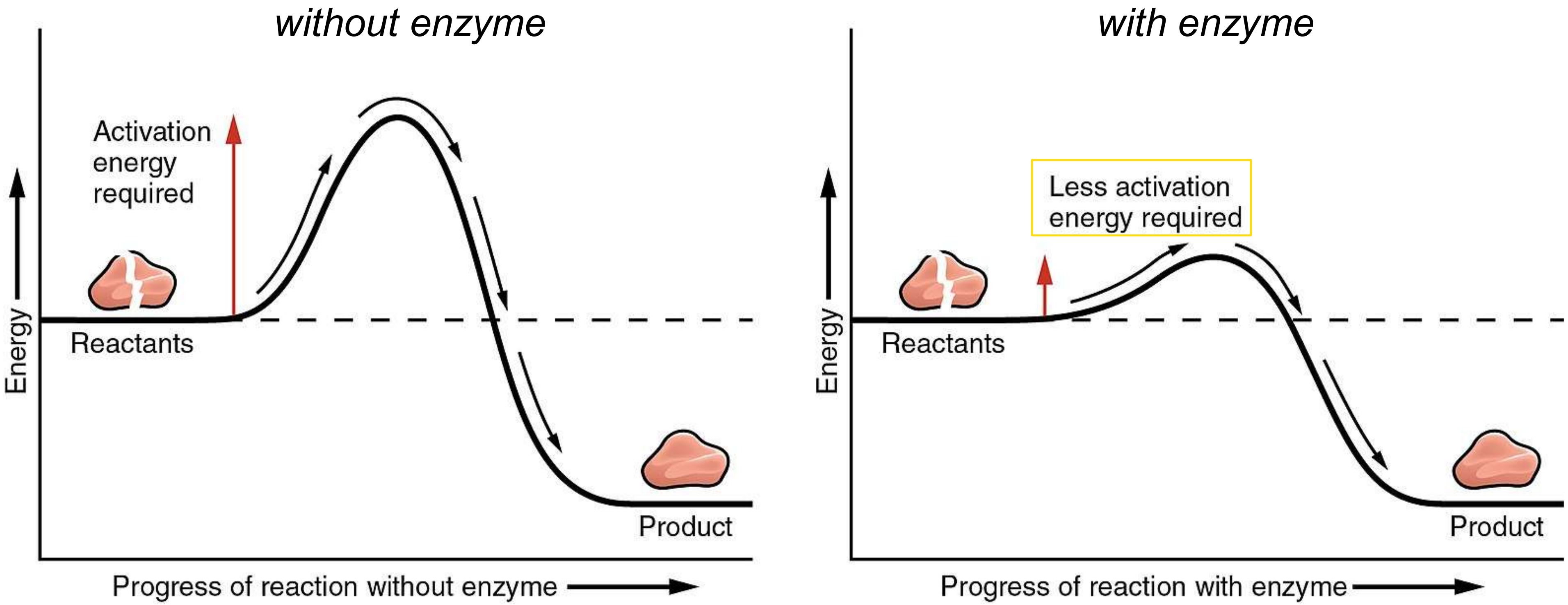


Figure: https://commons.wikimedia.org/wiki/File:212_Enzymes-01.jpg

- **Enzymes** speed up chemical reactions by <u>lowering the activation energy</u>
  - *Remember:* enzymes are proteins that are found associated with the cell membrane (to speed up chemical reactions there), but are also found in ALL parts of a cell, speeding up all chemical reactions

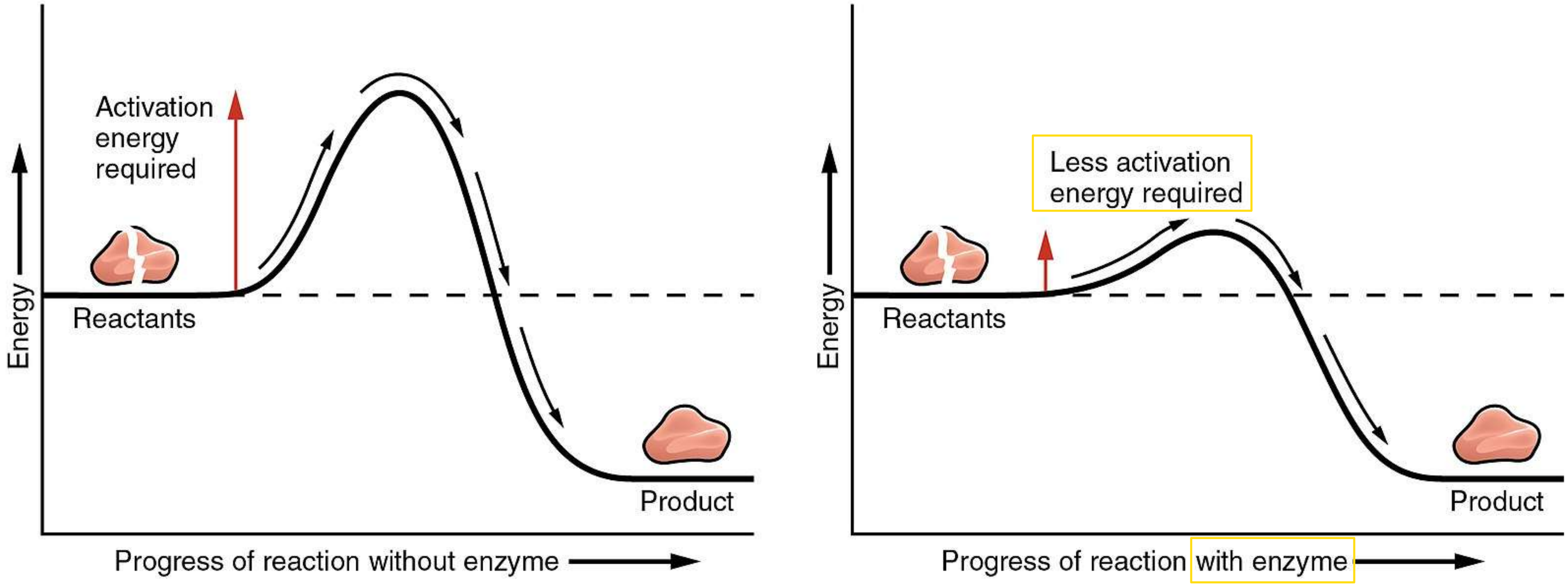


Figure: https://commons.wikimedia.org/wiki/File:212_Enzymes-01.jpg

Enzymes Key functions and properties

•**Catalysis:**
Enzymes accelerate chemical reactions that would otherwise occur too slowly to support life. They work by lowering the activation energy needed for a reaction.

•**Specificity:**
Each enzyme is highly specific and works on only one or a few types of molecules (substrates) to produce a specific product.

•**Regulation:**
Enzymes are vital for metabolism and can help build some substances and break down others.

•**Reusability:**
Enzymes are not used up in the reactions they catalyze and can be reused repeatedly.

Figure: https://commons.wikimedia.org/wiki/File:212_Enzymes-01.jpg

- *Remember*: all life requires energy
- *Remember*: to get energy, cells break apart glucose: exergonic chemical reaction
- *Remember*: to use the energy from glucose, cells store it in ATP: endergonic chemical reaction
  - <u>Therefore: life is dependent on chemical reactions</u>
- *Remember*: enzymes allow chemical reactions to occur fast enough to sustain life (up to 10^8-10^{20} times)
  - <u>Therefore: life is dependent on enzyme proteins</u>

Examples of Enzymes

- Digestive enzymes: Break down food into smaller molecules for absorption. Examples include amylase (carbohydrates), lipase (fats), and proteases (proteins).
- Ribozymes: A type of enzyme that is made of RNA instead of protein.
- Lysozyme: An enzyme found in secretions like tears and saliva that helps break down bacterial cell walls.
- DNA polymerase: An enzyme that synthesizes DNA molecules.

Figure: https://commons.wikimedia.org/wiki/File:212_Enzymes-01.jpg

Metabolism

Definition:

Metabolism refers to the set of life-sustaining chemical reactions that occur within living organisms

Functions:

1. Conversion of energy in food into a usable form for cellular processes;
2. Conversion of food to building blocks of macromolecules (biopolymers) such as proteins, lipids, nucleic acids, and some carbohydrates;
3. Excretion of metabolic wastes.

⬇

These enzyme-catalyzed reactions allow organisms to grow, reproduce, maintain their structures, and respond to their environments.

Metabolism

Metabolic reactions may be categorized as *catabolic*—the *breaking down* of compounds (e.g., breaking down glucose to pyruvate by cellular respiration);

or

anabolic—the *building up* (biosynthesis) of compounds (such as proteins, carbohydrates, lipids, and nucleic acids).

Usually, catabolism releases energy, and anabolism consumes energy.

- Life is dependent on chemical reactions & enzymes
  - The rate of chemical reactions in a cell = **metabolism**
    - high metabolism = many reactions, moving quickly
    - low metabolism = few reactions, moving slowly
  - Cells control metabolism by controlling enzymes
    - To speed up metabolism: make *more* enzymes & *activate* those that already exist
    - To slow down metabolism: make *less* enzymes & *de-activate* those that already exist

Enzymes in Drug Discovery

Enzymes are key components of metabolic pathways. Understanding how enzymes work and how they can be regulated is a key principle behind developing many pharmaceutical drugs

Digestive enzyme products may be given orally at meal times to improve digestion in people who cannot digest food properly because their pancreas does not produce the required amounts of enzymes (can occur as a result of cystic fibrosis, surgery, inherited conditions and other reasons).

Drug Name
lactase systemic Brand names: Lactaid, Lactaid Ultra, Lactrase, Lac-Dose, Lactaid Fast Act
pancrelipase systemic (Pro) Brand names: Zenpep, Creon, Pancreaze, Viokace, Pertzye
alpha-d-galactosidase systemic Brand names: Beano, Gas-X Prevention
sacrosidase systemic (Pro) Brand name: Sucraid
pancreatin systemic Brand names: Pan-2400, Pancreatin 4X, Hi-Vegi-Lip
cholic acid systemic (Pro) Brand name: Cholbam

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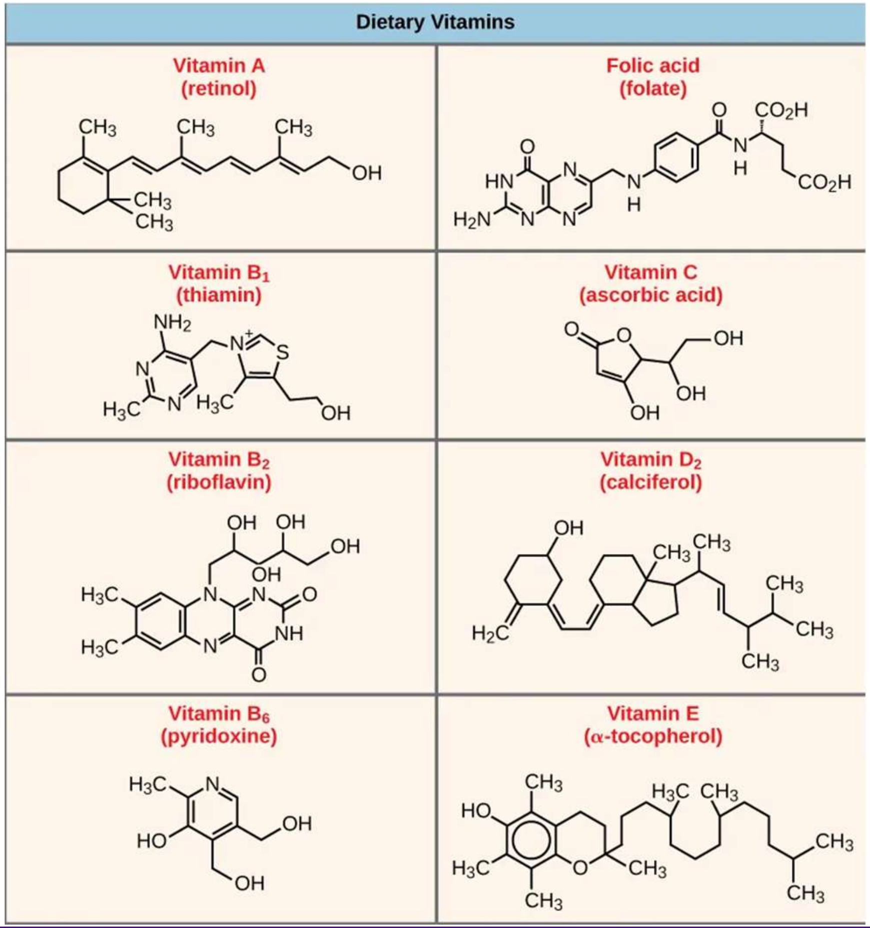
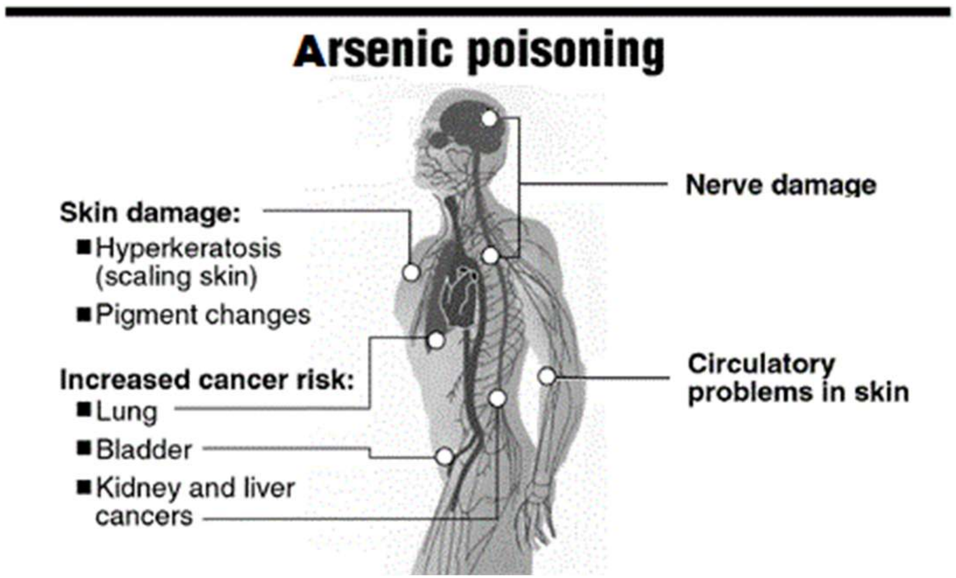


Co-enzymes

- Life is dependent on enzyme proteins: a problem with even one kind of enzyme can cause death
  - **Nerve gases** permanently block an enzyme responsible for muscle & brain activity: leads to paralysis & asphyxiation
    - e.g. Sarin gas attack in Syria, 2013

Figure: https://www.reuters.com/investigates/special-report/mideast-crisis-syria-chemicalweapons/

- <u>Life is dependent on enzyme proteins</u>: a problem with even one kind of enzyme can cause death
  - **Arsenic** blocks an enzyme used to make ATP
    - No energy = cell death
    - Often used in pesticides = present in our food, soil, & water



Arsenic poisoning to the skin is common in gold miners

Figure: http://article.sapub.org/10.5923.c.ije.201501.11.html
Figure: https://www.primehealthchannel.com/arsenic-poisoning.html

- <u>Life is dependent on enzyme proteins</u>: a problem with even one kind of enzyme can cause death

  – **Lead** blocks an enzyme that maintains the cell membrane

    • No membrane = cell death

    • Used in pesticides & found in paints, food, soil, & water, some cosmetics, & even toys

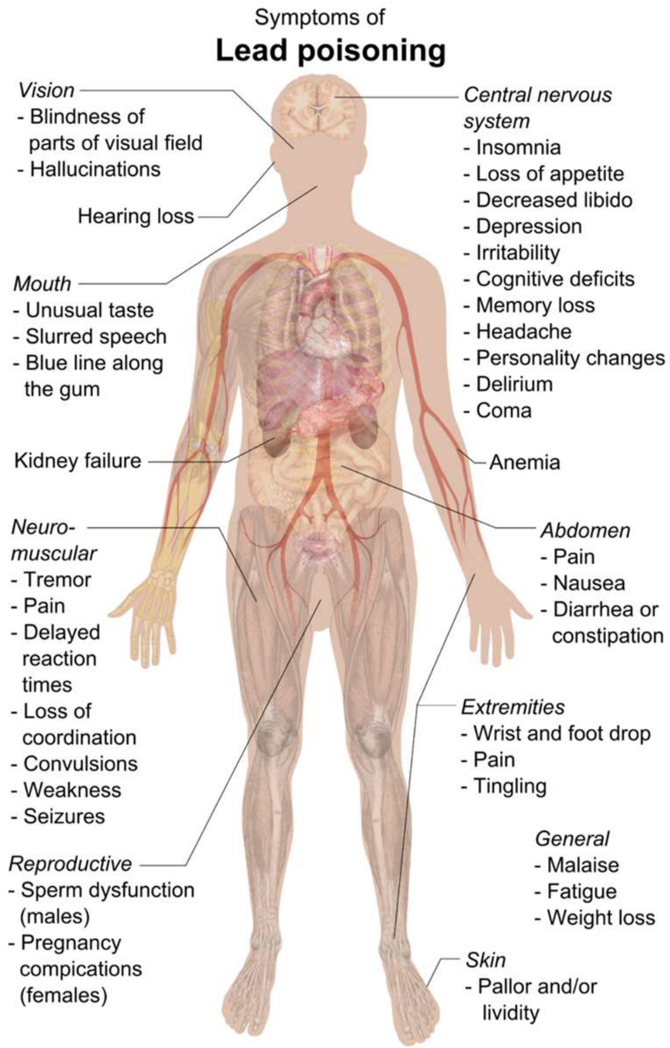


Figure: https://commons.wikimedia.org/wiki/File:Symptoms_of_lead_poisoning_(raster).png

- Remember: enzymes are proteins & all proteins must be folded into the correct structure to function
  - An unfolded protein = **denatured** protein = no longer functional
    - If our bodies get too hot, too acidic, too basic, too salty, etc., our enzymes will denature & we will die *(so we must maintain homeostasis)*

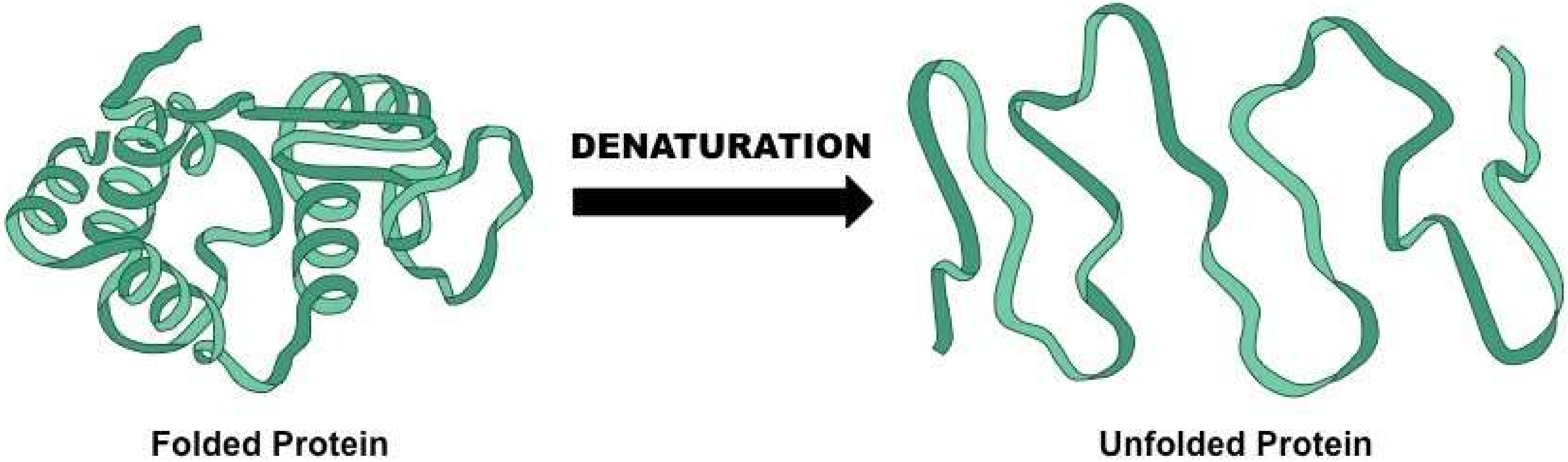


Figure: https://ib.bioninja.com.au/standard-level/topic-2-molecular-biology/24-proteins/denaturation.html

- <u>The surrounding environment, such as the pH & the temperature, affects enzymes</u>
  - Changing the pH to be more acidic or basic can cause enzyme denaturation
    - *must maintain constant pH of the blood (homeostasis)*

For most enzymes, max activity occurs at about pH 7.4 (close to neutral), though some work better at basic pH and some at acidic (such as stomach enzymes)

fast

enzyme
activity

slow

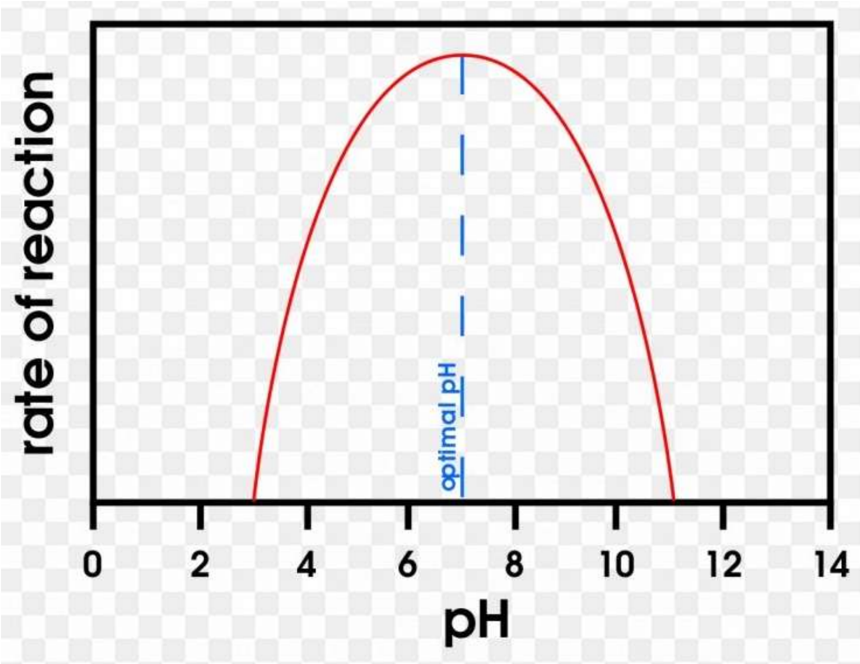


Figure: https://favpng.com/png_view/do-the-old-effect-ph-enzyme-denaturation-thermodynamic-activity-reaction-rate-png/DcheKRLW

- <u>The surrounding environment, such as the pH & the temperature, affects enzymes</u>
  - – Changing temperatures to be cooler slows down enzymes
  - – Changing it to be hotter can cause denaturation
    - *must maintain constant body temperatures (homeostasis)*

For most human enzymes, maximum activity occurs at about 98.6°F (37°C)

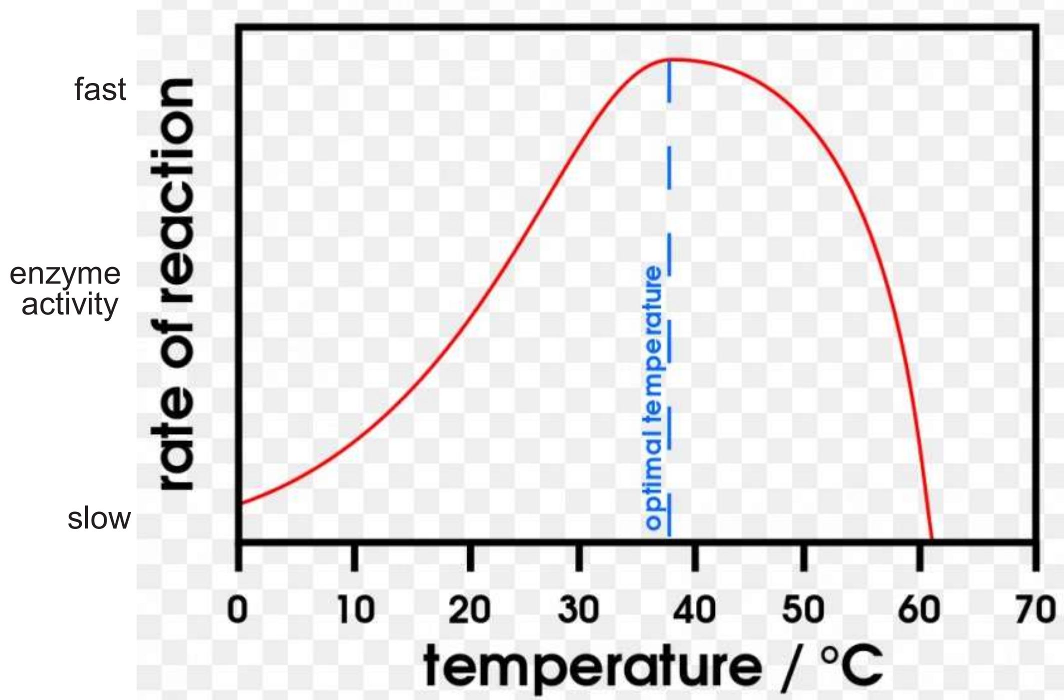
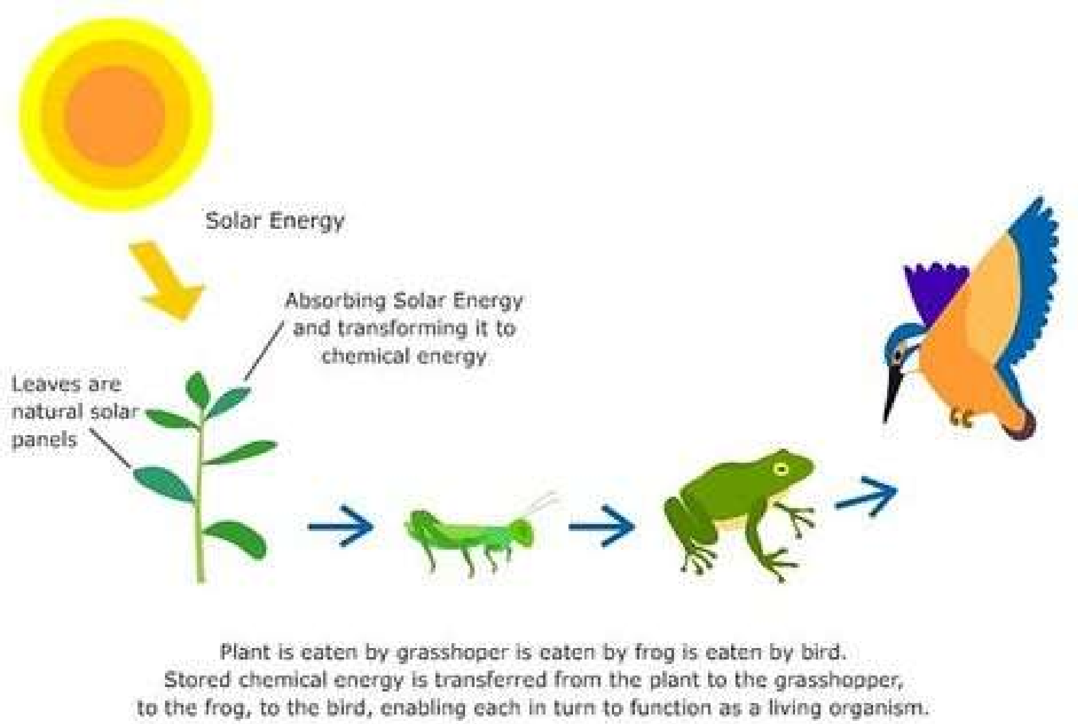


Figure: https://favpng.com/png_view/enzyme-substrate-chemical-reaction-thermodynamic-activity-ph-png/ZjBaMNGC

How do we get usable energy on Earth?

- Chemical reactions store or release energy: so how do we get the energy on Earth in the first place?
  - The most important endergonic reaction for life = **photosynthesis**
    - energy for living things on Earth comes from the sun



- **Photosynthesis** = process by which some organisms capture sunlight & store it in the bonds of a sugar molecule
  - plants, protists (algae), & some bacteria perform photosynthesis
    - life on Earth depends on these organisms

Figure: https://commons.wikimedia.org/wiki/File:CSIRO_ScienceImage_4628_Bluegreen_algae_in_irrigation_drain.jpg

- Most organisms that undergo photosynthesis appear green
  - *Remember*: **chloroplasts** are the organelles inside cells where photosynthesis takes place
    - Chloroplasts contain light-capturing pigments (to trap sunlight)
    - Most common pigment = **chlorophyll**, which is green

The photosynthetic cells of a leaf are packed with chloroplasts, which contain green chlorophyll

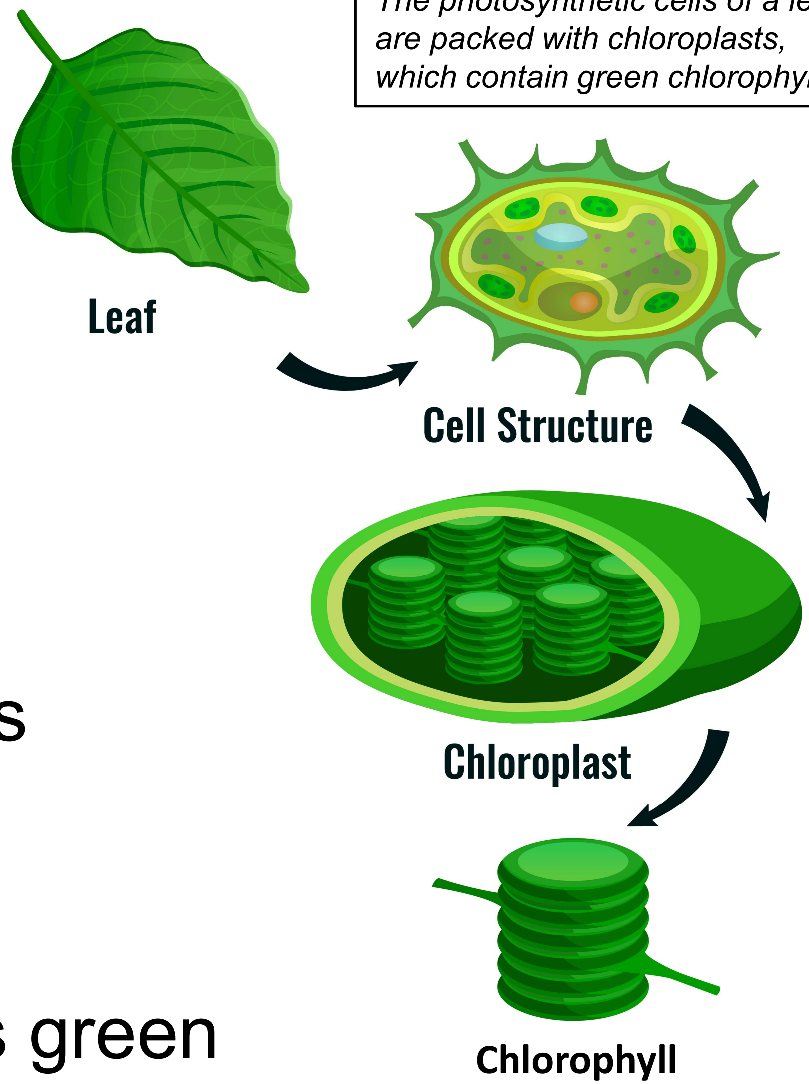
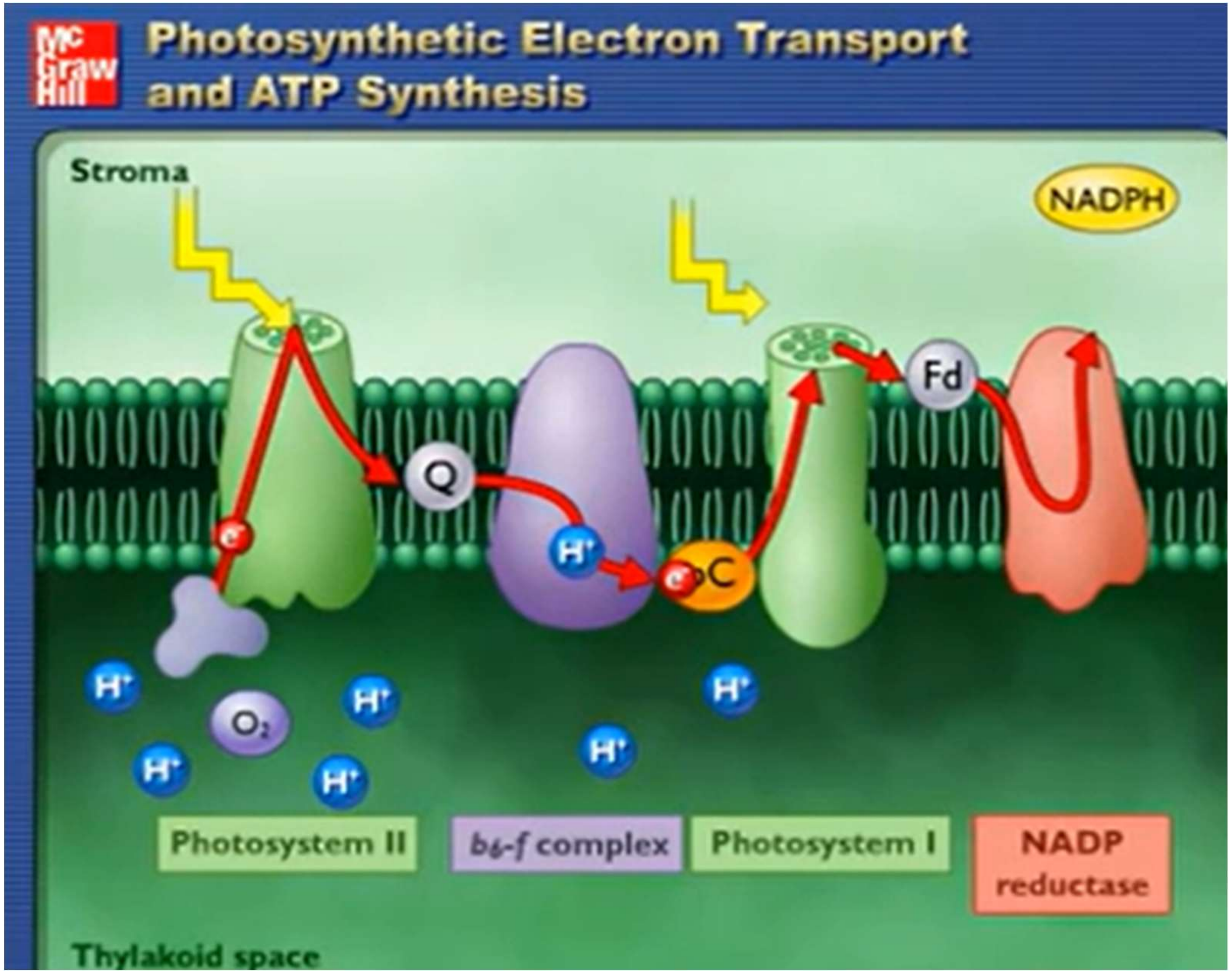


Figure: https://www.freepik.com/free-vector/chloroplast-converting-light-energy-into-sugar-green-plant-cells-food-photosynthesis-infographic-elements-schema_6847346.htm

Photosynthetic Electron Transport and ATP Synthesis



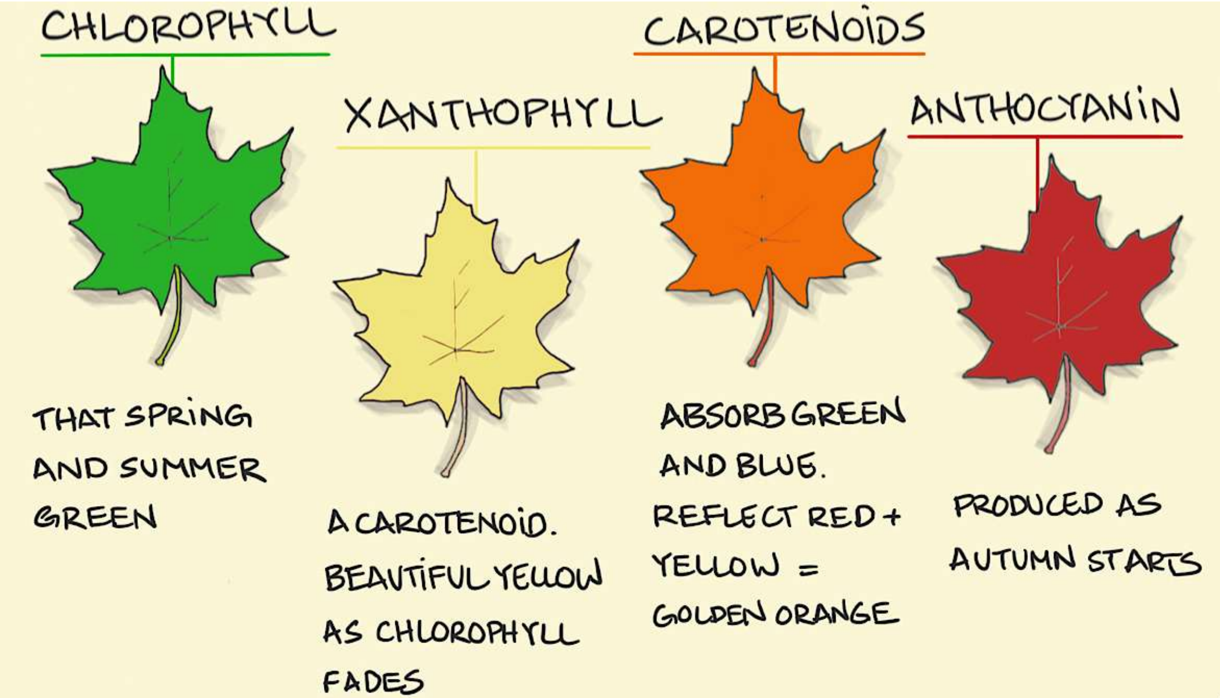
https://www.dailymotion.com/video/x2z1u7y

- If chlorophyll is green, why are some plants other colors?
  - Chlorophyll is the most common pigment, but plants have many other pigments
    - e.g. carotenoids (red, orange, & yellow)
    - e.g. anthocyanins (blue or purple)
      - Some plants have more carotenoids or anthocyanins, so they're purple or red, etc.

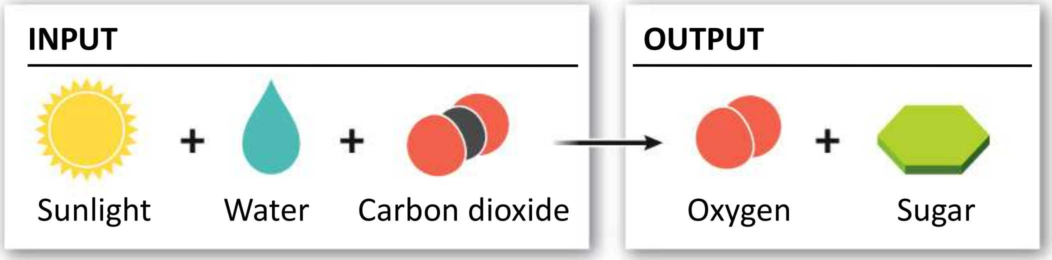
Figure: https://www.piqsels.com/en/public-domain-photo-fblzm

Figure: https://www.flickr.com/photos/mercadanteweb/8170759767

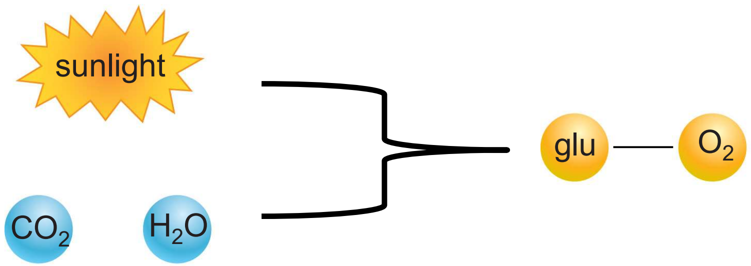
- Most plants have many types of pigments, but more of the chlorophyll in spring & summer – look green
  - When trees start to go dormant in the fall, chlorophyll is the 1st pigment to go, leaving other pigments to show through



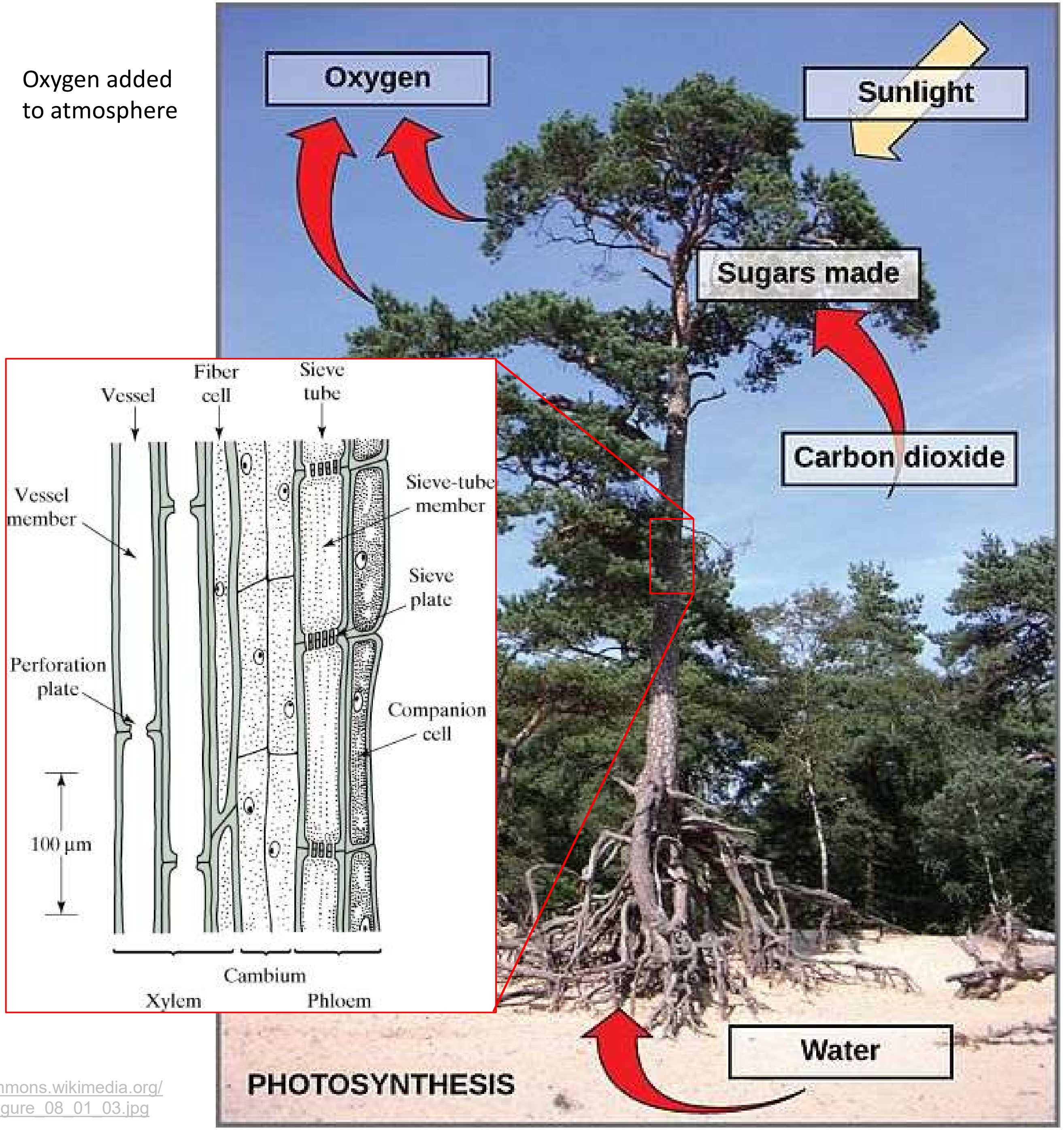
- How do plants capture sunlight & turn it into sugar/stored energy?
  – Sunlight is captured in pigments, then combined with water & CO_2 to produce a sugar molecule
  – O_2 is given off as waste



  – *Remember: photosynthesis stores energy: endergonic*



Oxygen added to atmosphere



Energy from sun captured and stored in sugar molecules

Sugar = energy used to grow plant & keep it alive

Carbon dioxide absorbed from atmosphere

Water absorbed from ground through roots

Figure: https://commons.wikimedia.org/wiki/File:Figure_08_01_03.jpg

- <u>Gas exchange: plants take in CO_2 & give off O_2</u>
  - When humans grow, new tissue & cells are built from the food we eat
  - When plants grow, where does the new tissue come from?
    - not from the soil: as a plant grows, the soil under it doesn't disappear
    - not from the water: bigger plants contain more water, but it doesn't account for the overall increase in solid matter
    - comes from the air! – CO_2

Carbon stays, building the molecules in the plant

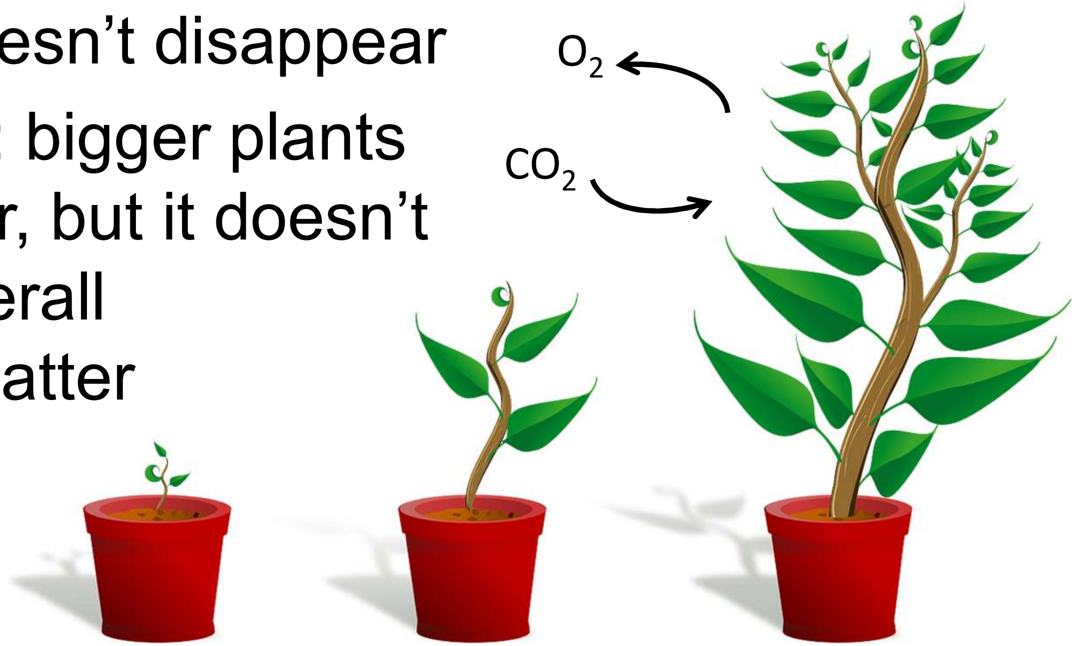
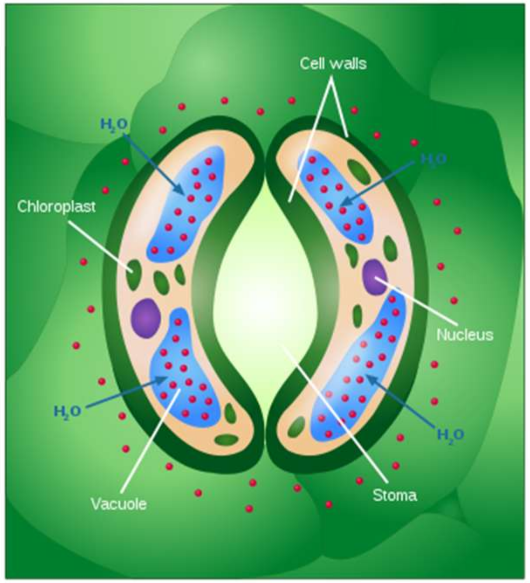
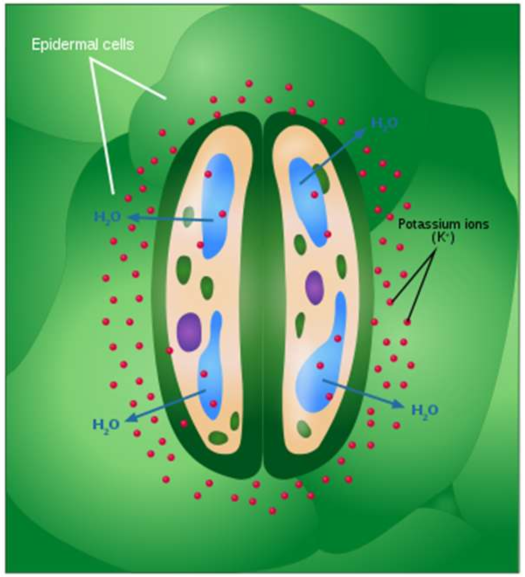


Figure: https://pixabay.com/vectors/sapling-plant-growing-seedling-154734/

- <u>Plants must constantly exchange gases (CO_2 & O_2) to stay healthy</u>
  - But they don't have lungs
  - Use pores in leaves instead = **stomata**
    - when stomata open, CO_2 comes in & O_2 goes out, but water also evaporates out
      - stomata close to prevent too much water loss



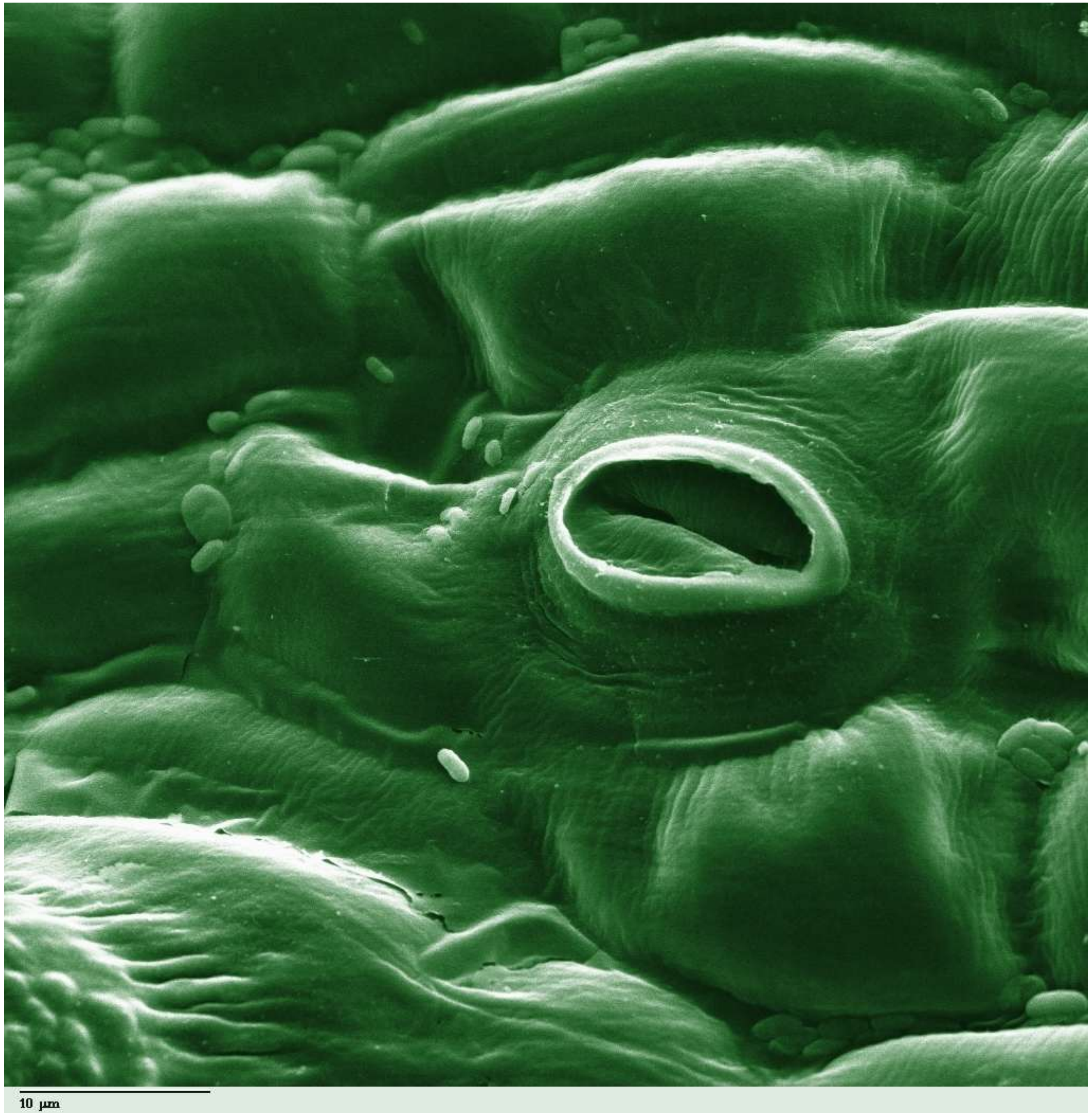
Stoma opening



Stoma closing

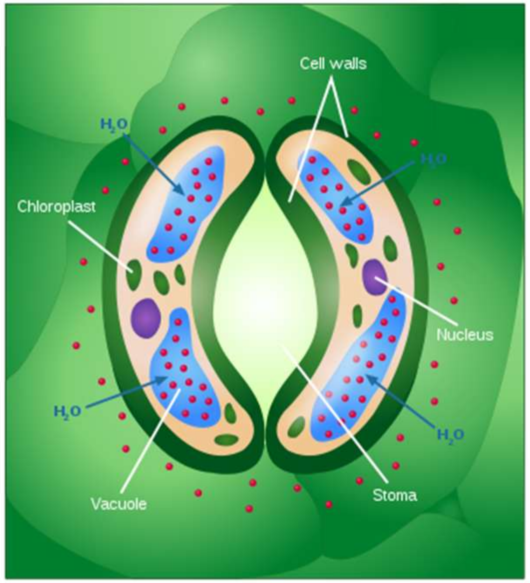
Figure: https://commons.wikimedia.org/wiki/File:Opening_and_Closing_of_Stoma.svg

- **stoma**

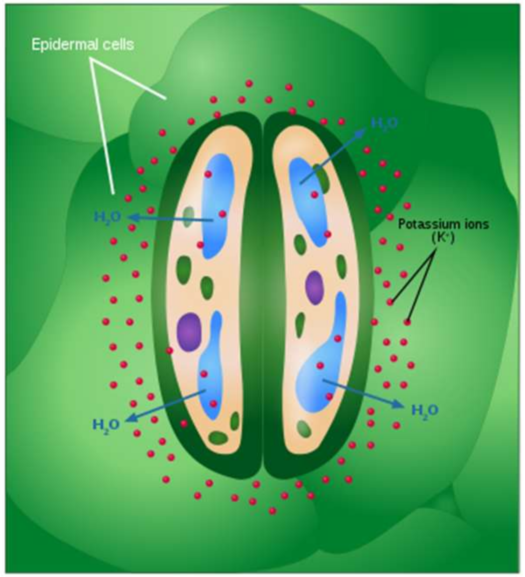


Tomato leaf stomata

- Leaves are always compromising: open stomata just enough to exchange gases, but not enough to lose too much water
  - In hot or dry environments, stomata stay closed more often to prevent water loss
  - BUT, this allows O_2 to build up inside the plant cells which can be toxic = **photorespiration**
    - **Photorespiration** = the build-up of O_2 to toxic levels: can eventually lead to death



Stoma opening



Stoma closing

Figure: https://commons.wikimedia.org/wiki/File:Opening_and_Closing_of_Stoma.svg

- Plants in hot or dry environments are faced with a hard choice:
  - 1. open stomata & dehydrate (can = death)
  - 2. keep stomata closed & risk photorespiration (can = death)

C_3 plants can die of dehydration or photorespiration in hot or dry environments

- The average plant is called a **C_3 plant**
  - the majority of plants on Earth are C_3 plants
  - these are at risk in hot or dry environments, because they are susceptible to photorespiration

Figure: https://commons.wikimedia.org/wiki/File:Starr-100505-6018-Senecio_madagascariensis-dead_brown_plants-Polipoli-Maui_(24407087694).jpg

- Some plants evolved a mechanism to prevent photorespiration
  - **C_4 plants** use special cells to minimize photorespiration
    - e.g. corn & crabgrass
  - **CAM plants** open stomata based on time of day
    - e.g. cacti & pineapple
      - At night stomata are open: minimal water loss
      - During the day stomata are closed: less water loss

Figure: https://pixabay.com/photos/corn-plant-corn-on-the-cob-896665/

Figure: https://www.flickr.com/photos/75885098@N05/50732629663

- If C_4 & CAM plants don't have to worry about photorespiration, why haven't more plants evolved to use these strategies?
  - C_4 & CAM plants use more energy than C_3 plants
    - they ONLY have an advantage in very hot or dry environments
  - Where water is abundant or light levels are low, C_3 plants have the advantage
    - they do not have to use so much energy for normal functions

C_3 plants use less energy

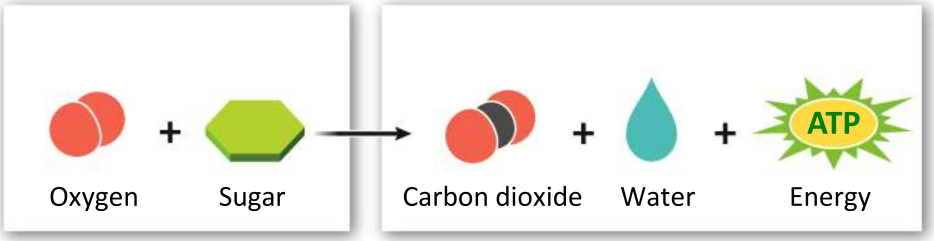
C_4 & CAM plants need more energy to function

Figure: https://pixnio.com/flora-plants/flowers/roses-flower-pictures/leaf-flora-nature-flower-rose-plant-dew-raindrop#

Figure: https://www.flickr.com/photos/75885098@N05/50732629663

How do we release energy trapped in molecules?

- Once photosynthetic organisms have stored energy in the bonds of sugar, how do we release the energy in order to use it?
  - The most important exergonic reaction for life = **glucose breakdown**



    - All living things (plants, animals, fungi, bacteria, etc.) must breakdown glucose molecules to release energy to stay alive

- How do organisms get glucose?
  - – Plants & other photosynthetic organisms make it by themselves from sunlight
  - – Animals & other organisms eat plants (or organisms that ate plants)
    - also get it from other organic molecules (lipids, proteins, etc.) that we can convert to glucose

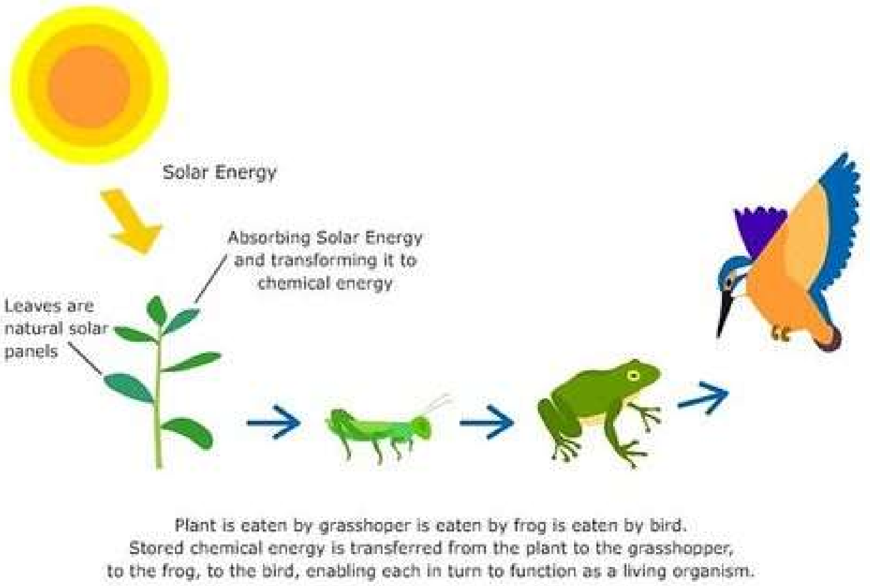
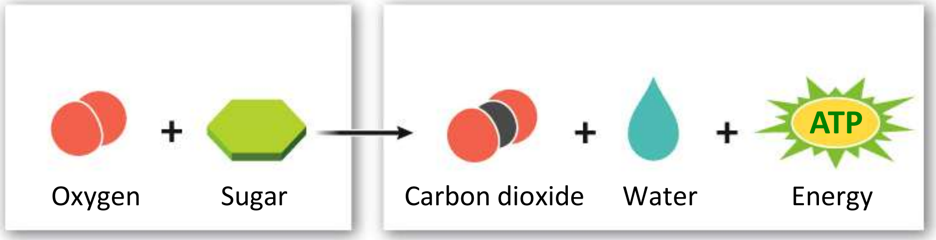


Figure: https://www.sciencelearn.org.nz/images/2174-a-food-energy-chain

- *Remember*: the energy in glucose can't be used directly – must first transfer it to an energy-carrier molecule like **ATP**
  - we can then use ATP directly



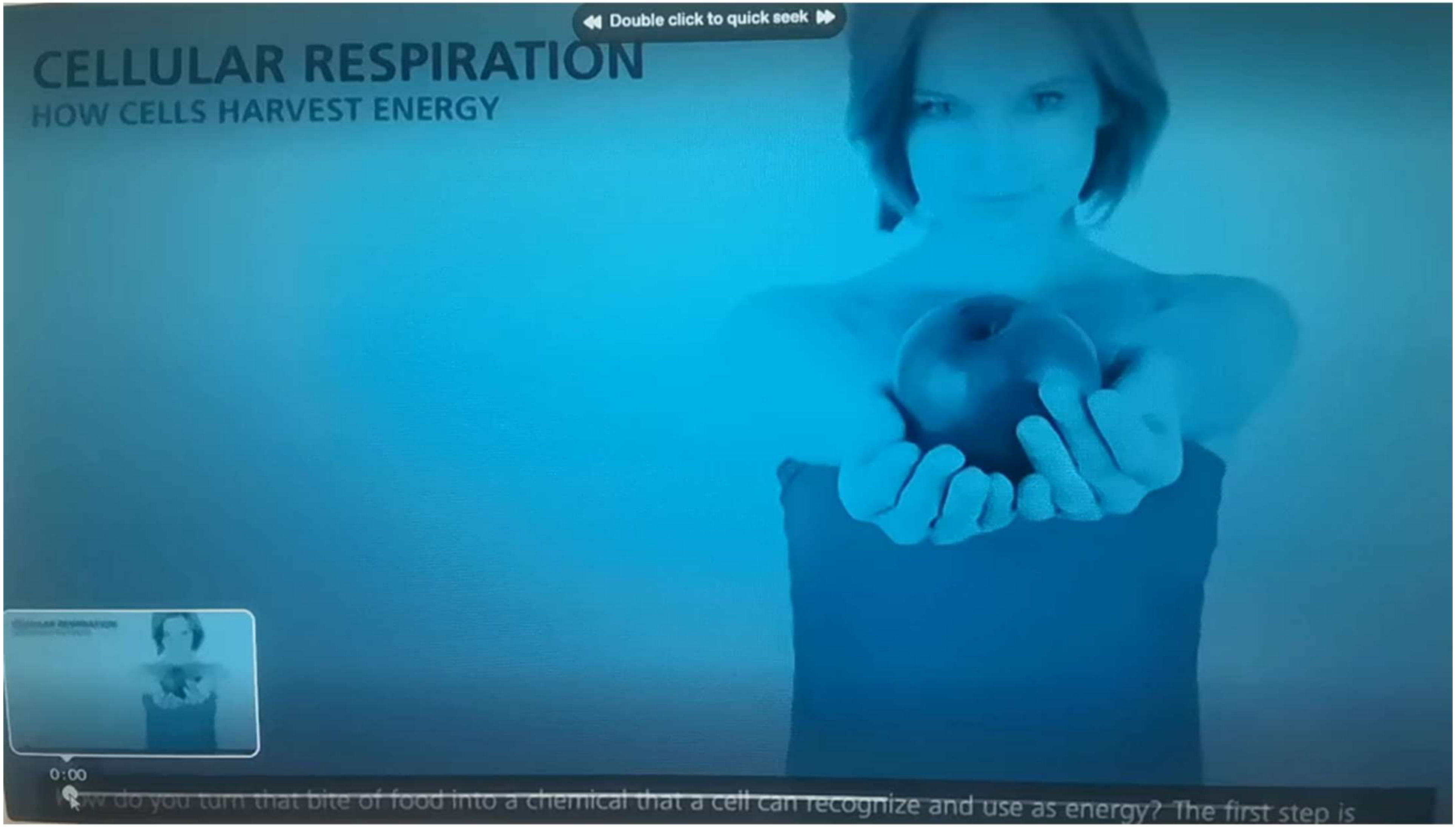
- Glucose breakdown happens in 2 steps:

Step 1 = **glycolysis** *(in the cytoplasm of cells)*

If O_2 is available → Step 2 = **cellular respiration** *(in the mitochondria)*

If **no** O_2 available → Step 2 = **fermentation** *(in the cytoplasm)*

- <u>Cellular Respiration</u>

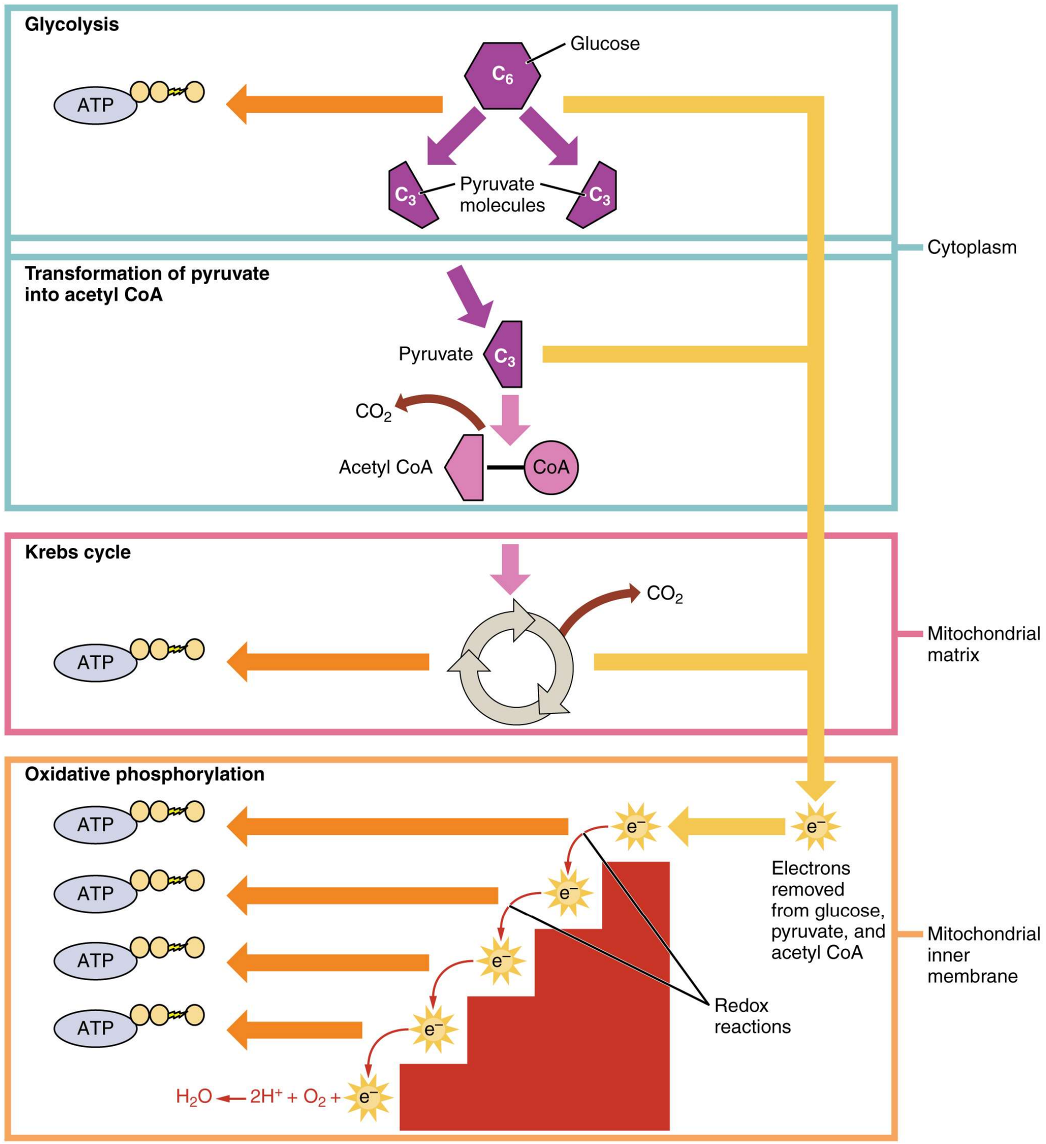


https://www.dailymotion.com/video/x7tl5p9

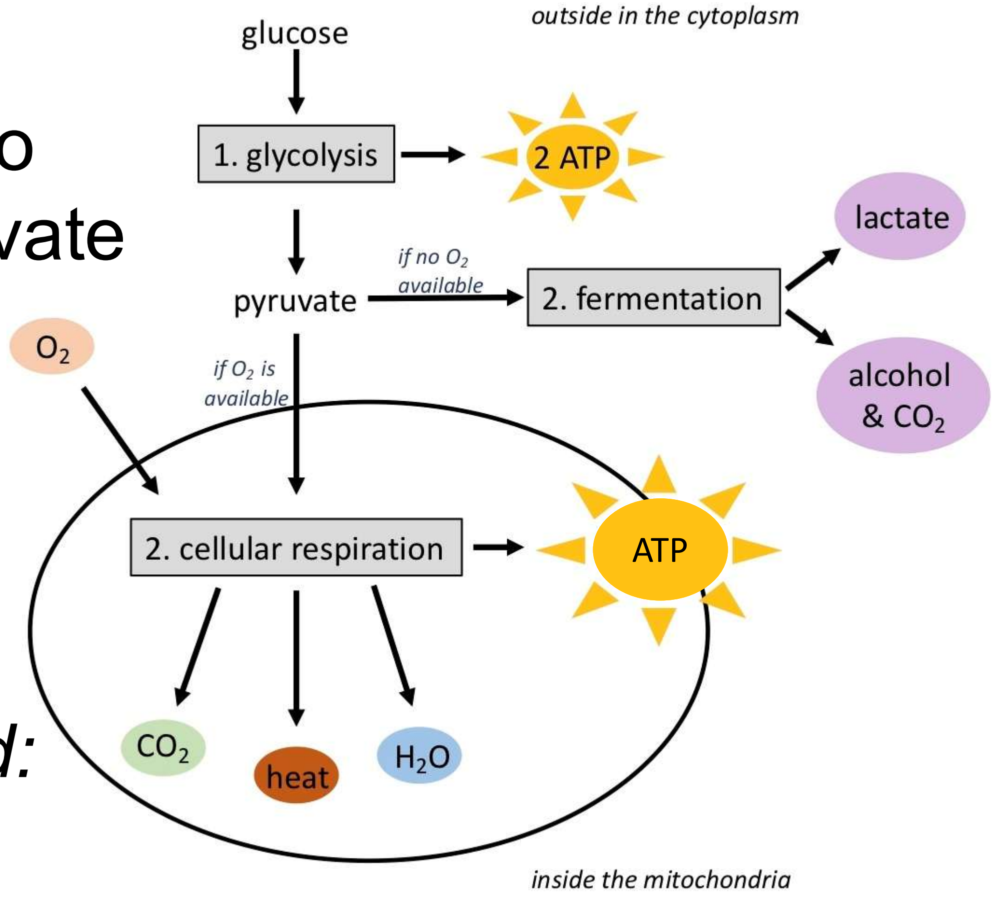
Cellular Respiration

Cellular respiration is the process of oxidizing biological fuels using an inorganic electron acceptor, such as oxygen, to drive production of adenosine triphosphate (ATP), which stores chemical energy in a biologically accessible form.

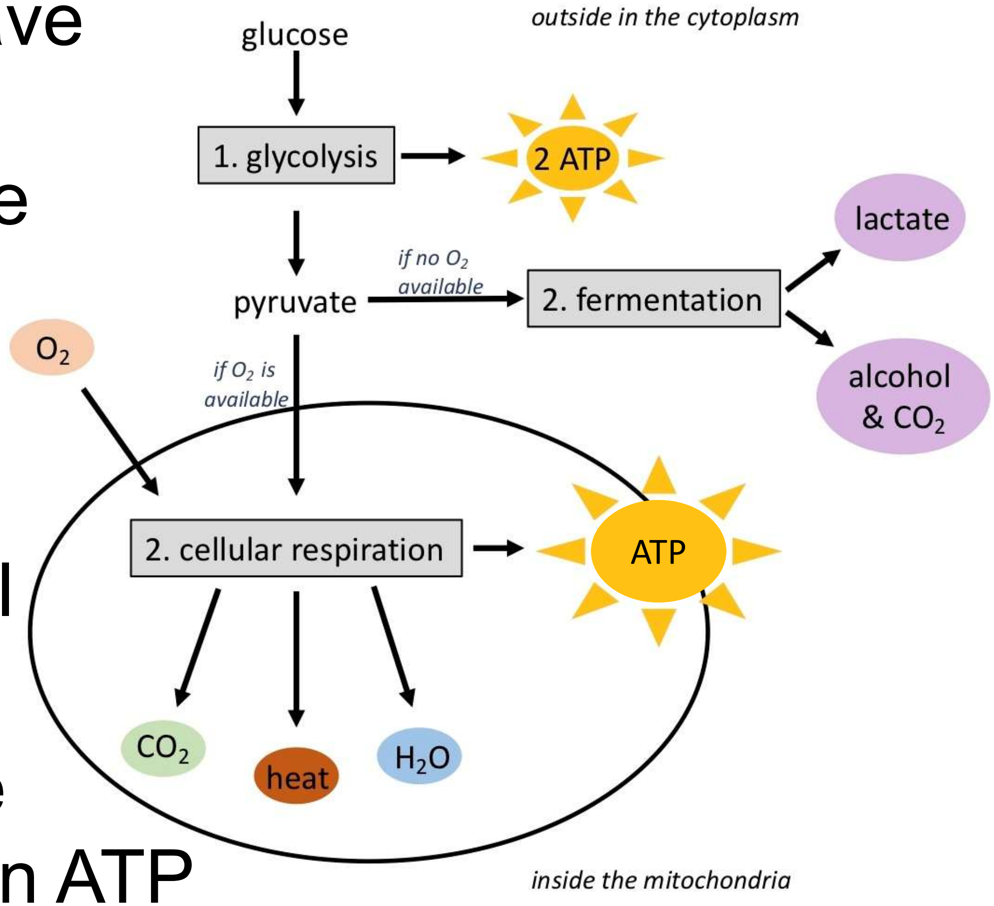
Cellular respiration may be described as a set of metabolic reactions and processes that take place in the cells to transfer chemical energy from nutrients to ATP, with the flow of electrons to an electron acceptor, and then release waste products



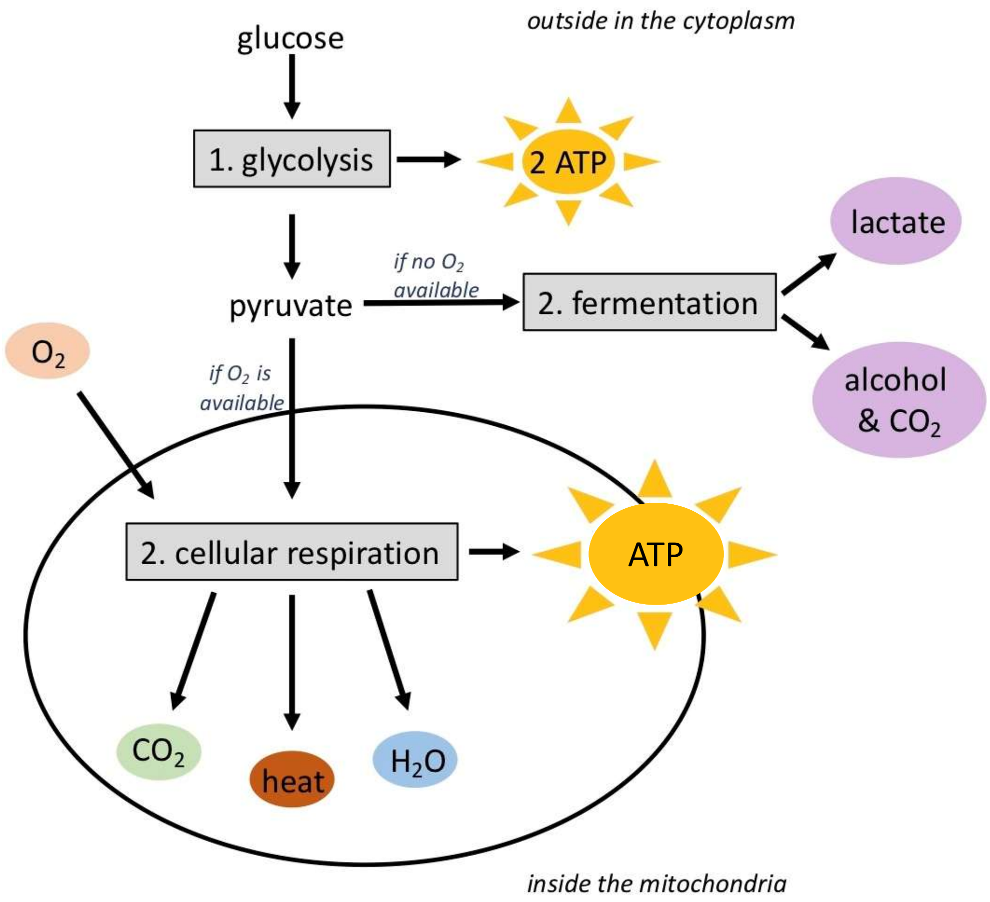
- Step 1 is always **glycolysis** *(occurs in cytoplasm)*
  - glucose → 2 pyruvate & 2 ATP
  - breaks the 1st bond in glucose, causing the molecule to break into 2 pieces: called pyruvate
    - releases 2 ATPs (energy) by breaking the bond
  - *only glucose required: no O_2 needed*



- 2 ATPs is not very much & there are still many more bonds to break
  - If O_2 is available, pyruvate will get shipped into the mitochondria to have the rest of their bonds broken & to give us the most ATP
  - *Remember*: the **mitochondria** is the powerhouse of the cell
    - its main job is to break apart glucose & store the energy in ATP

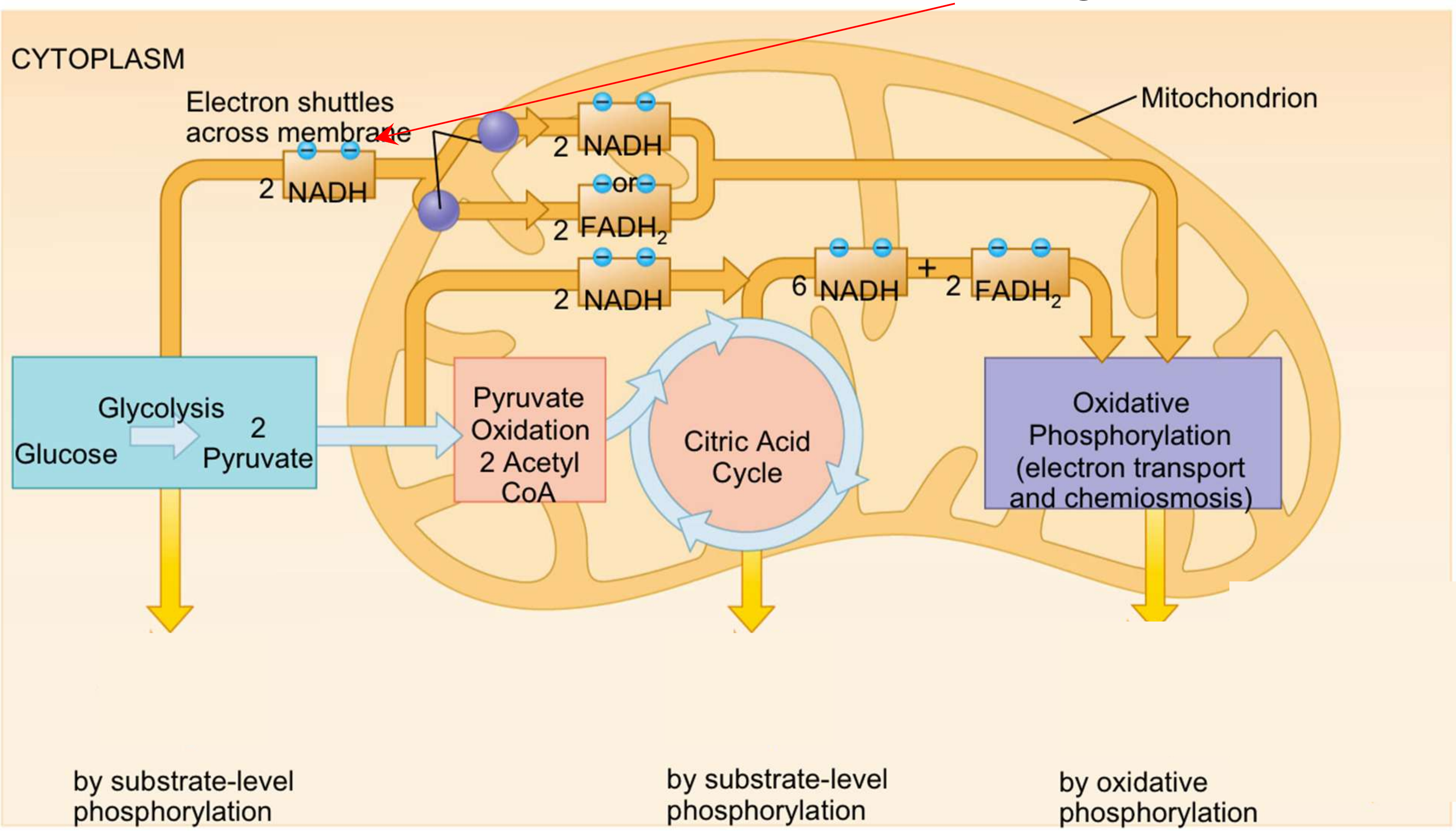


- Step 2 if O_2 is available = **cellular respiration**
  - Pyruvate moves into the mitochondria, which also pulls in O_2
  - Because cellular respiration requires O_2 it is called **aerobic**
    - why we must breathe!



Energy Yield from the Complete Oxidation of One Molecule of Glucose

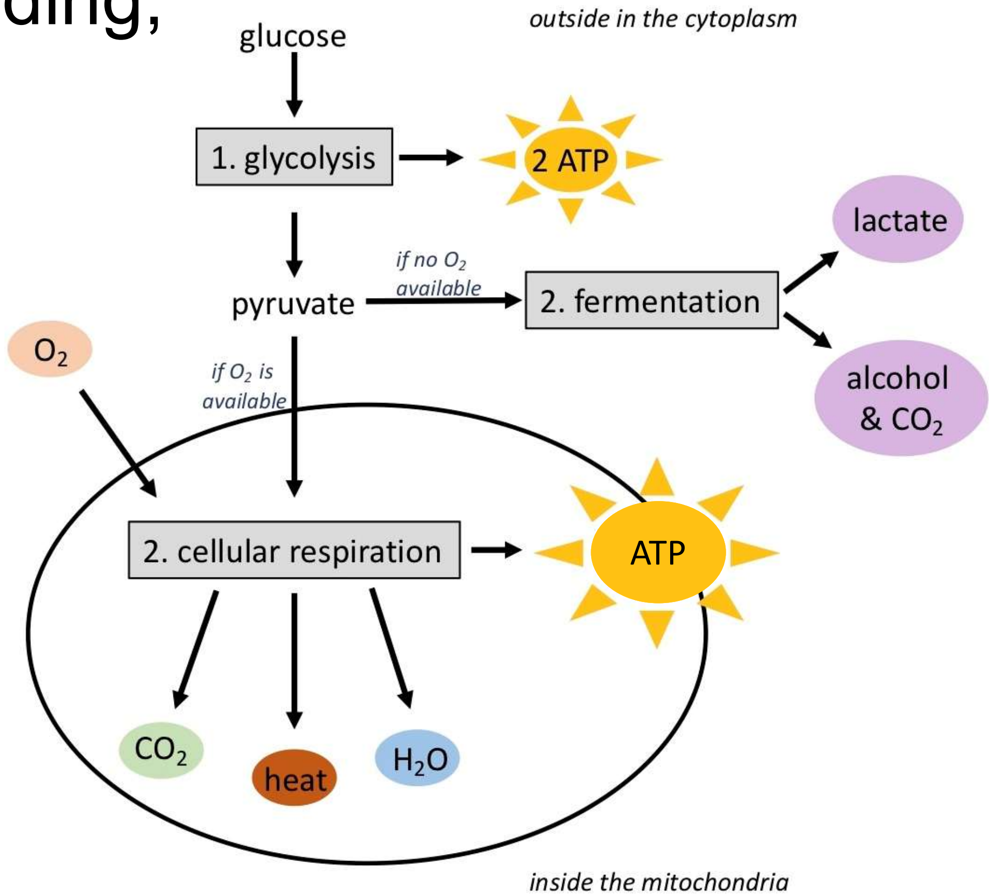
- 1 NADH $\rightarrow$ 2.5 ATP, 1 $FADH_2$ $\rightarrow$ 1.5 ATP
- For some cells, minus 2 when transporting these 2 NADH.



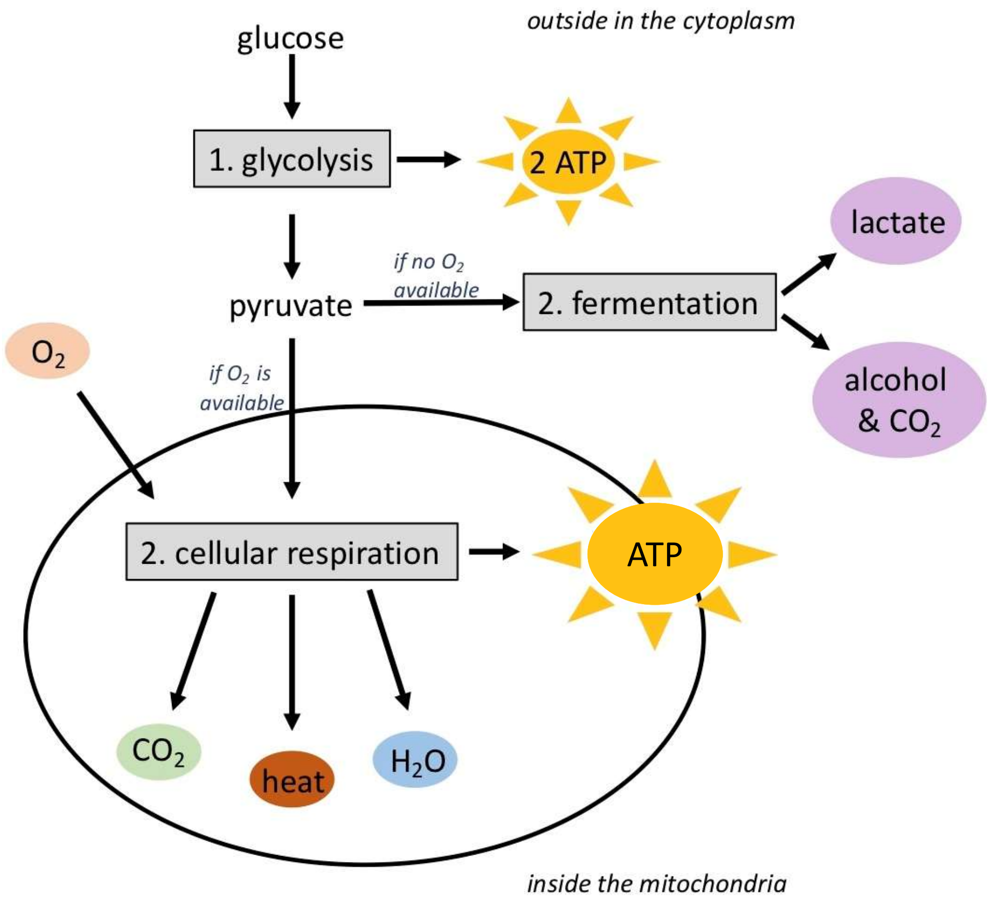
- **pyruvate + O_2 → 25 ATP + CO_2 + H_2O + heat**

– During cellular respiration 30 ATP are created (including, using the 2 NADH from glycolysis) depending on the efficiency of the cell

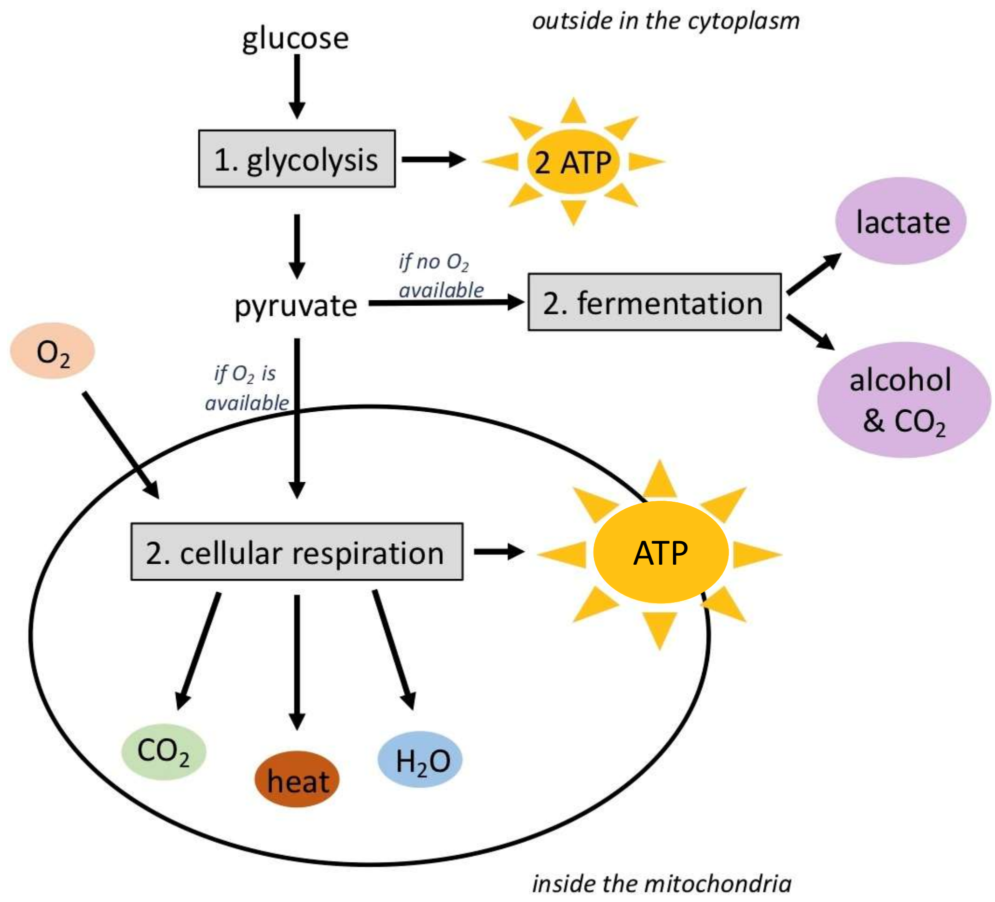
– *Remember*: exergonic reactions are inefficient, we will always have a byproduct of heat



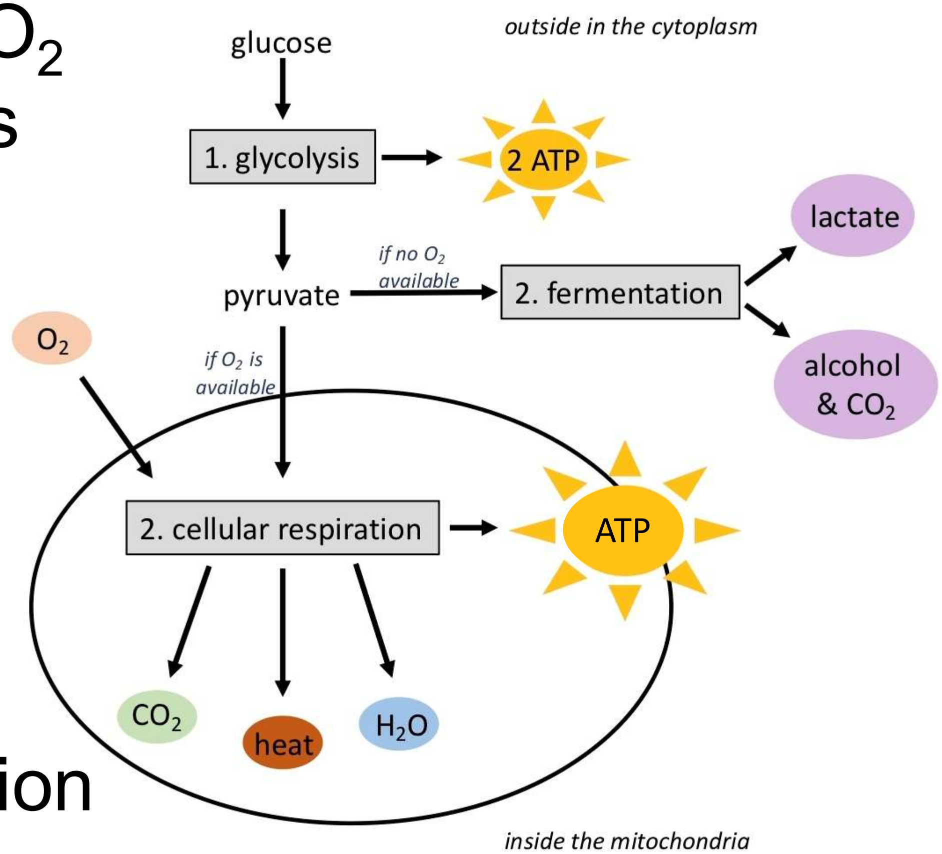
- <u>When O_2 is available:</u>
  - we got 2 ATP from glycolysis
  - we got 30 ATP from cellular respiration
  - so for each molecule of glucose we break down, we get a total of **30-32 ATP**



- Step 2 if O_2 is NOT available = **fermentation**
  - – e.g. muscles that are working hard use up O_2 very quickly → low O_2 = fermentation
  - – if no O_2, glycolysis still occurs (2 ATP), but pyruvate is not shipped into the mitochondria
  - – instead, pyruvate stays in cytoplasm & goes through fermentation



- **Fermentation** makes no ATP
  - Instead, turns pyruvate into:
    - lactate (lactic acid) if in animal cells & some bacteria
    - alcohol (ethanol) & CO_2 if in yeast & plant cells
  - a small amount of fermentation is good: necessary to regenerate certain molecules used for cellular respiration



- **Lactic acid fermentation in animals**
  - When this builds up in muscles that are working hard, it can cause burning
    - Cleared very quickly, but athletes are experiencing this at certain times when they are "feeling the burn"

Figure: https://www.shutterstock.com/image-photo/sore-muscles-legs-caused-by-exercise-1376564054

- **Fermentation to make food products**
  - – Lactic acid fermentation in bacteria is used to make foods like sauerkraut & yogurt
  - – Alcohol fermentation in yeast is used to make foods like beer, wine, & bread

Figure: https://www.flickr.com/photos/stone-soup/14852461299

Figure: https://pixnio.com/food-and-drink/beer/beer-and-bread

- A small amount of fermentation in cells is good, but we still need cellular respiration to function: need all that ATP energy from glucose to stay alive
  - Blocking cellular respiration = unconsciousness, brain damage, & within minutes can cause death
    - e.g. stopping breathing/suffocation *(cellular respiration needs O_2 to make ATP)*
    - e.g. being poisoned with **cyanide** *(prevents O_2 from being used in the mitochondria)*
      - fastest acting poison: within minutes, just like suffocation

Lethal dose of cyanide

Figure: https://kochipost.com/2019/10/17/the-effects-of-cyanide-on-the-human-body/

- Summary

- Energy & bioenergy
- Exergonic and endergonic reactions
- Metabolism & enzymes
- Energy in cells
- Cellular Respiration

Review Q to help your understanding

1.What is the purpose of cellular respiration. Provide a concise summary of the process.

2.Draw and explain the structure of ATP (Adenosine Tri-Phosphate).

3.State what happens during glycolysis

4. Describe the structure of a mitochondrion.

5.What happens during the electron transport stage of cellular respiration

6. How many molecules of ATP can be produced from one molecule of glucose during all three stages of cellular respiration combined?

7.Do plants undergo cellular respiration, Why or why not?

8.Explain why the process of cellular respiration described in this section is considered aerobic

9. Name three energy-carrying molecules involved in cellular respiration

10..*True or False*. The carbons from glucose end up in ATP molecules at the end of cellular respiration

11.Which stage of aerobic cellular respiration produces the most ATP?